

Machine Center Base [FB2]

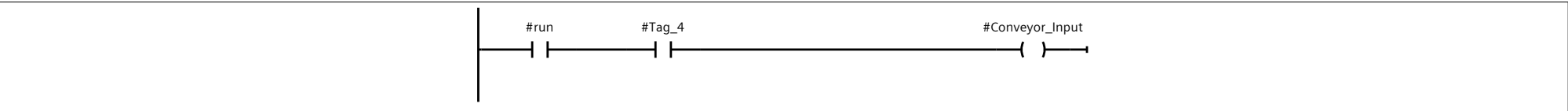
Machine Center Base Properties							
General							
Name	Machine Center Base	Number	2	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name		Data type	Default value	Retain
▼ Input				
FCIO		Bool	false	Non-retain
SensorLidsIntrare		Bool	false	Non-retain
SensorLidslesire		Bool	false	Non-retain
Machine_Error		Bool	false	Non-retain
Machine_Busy		Bool	false	Non-retain
PanelStopButton		Bool	false	Non-retain
PanelResetButton		Bool	false	Non-retain
run		Bool	false	Non-retain
▼ Output				
Conveyor_Input		Bool	false	Non-retain
Start_Machine		Bool	false	Non-retain
Counter		DInt	0	Non-retain
Conveyor_Output		Bool	false	Non-retain
Stop_MachineCenter		Bool	false	Non-retain
Reset_MachineCenter		Bool	false	Non-retain
InOut				
▼ Static				
IEC_Counter_0_Instance		CTU_INT		Non-retain
IEC_Timer_0_Instance		TON_TIME		Non-retain
IEC_Timer_0_Instance_1		TON_TIME		Non-retain
Tag_3		Bool	false	Non-retain
Tag_4		Bool	false	Non-retain
orice		Bool	false	Non-retain
IEC_Counter_0_Instance_1		CTU_INT		Non-retain
IEC_Timer_0_Instance_2		TP_TIME		Non-retain
IEC_Timer_0_Instance_3		TON_TIME		Non-retain
▼ Temp				
Tag_5		Bool		
orice1		Bool		
Constant				

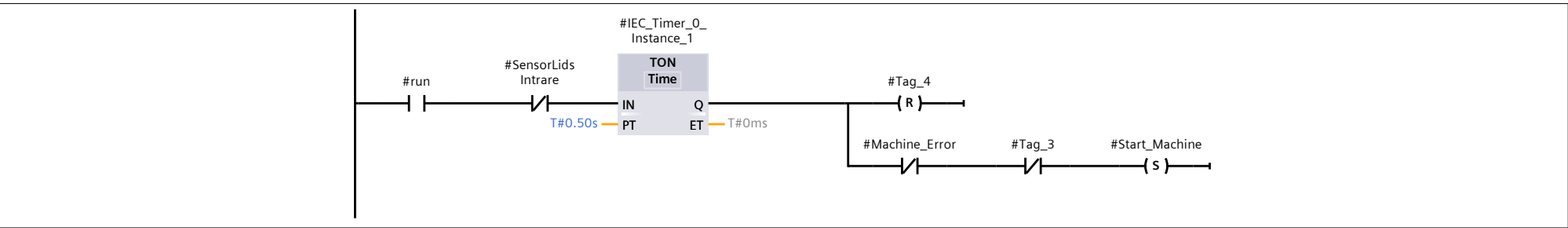
Network 1:



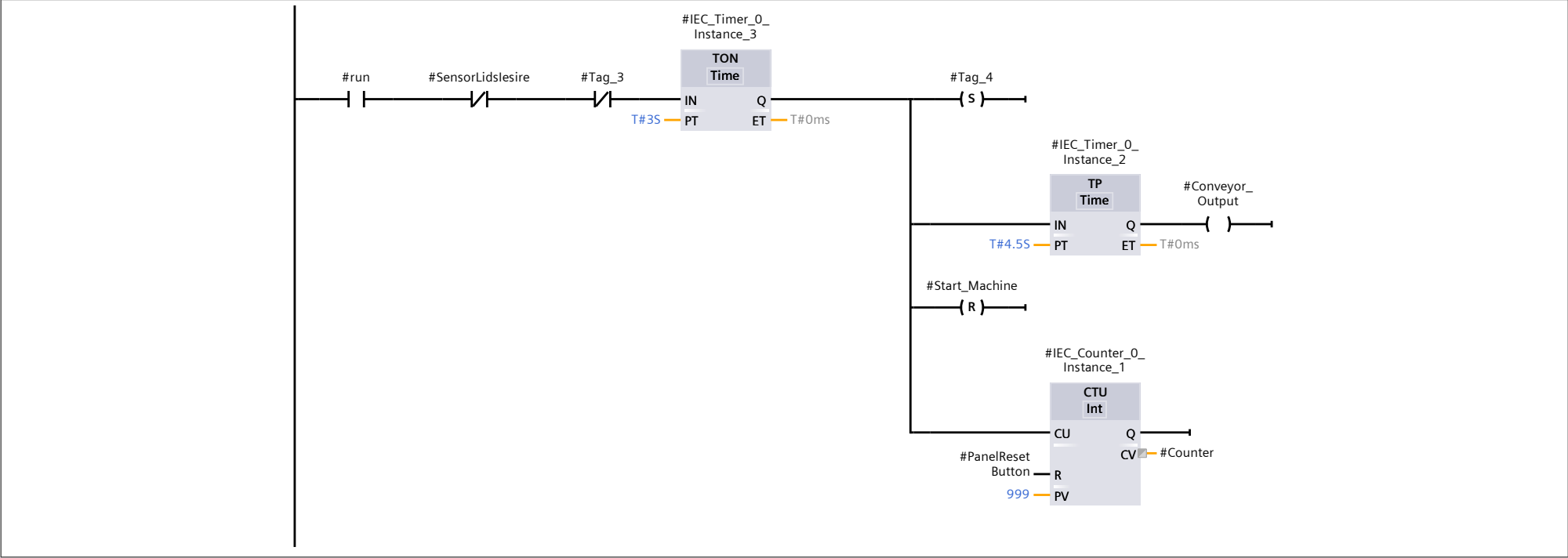
Network 2:



Network 3:



Network 4:



Network 5:



Totally Integrated Automation Portal

Assembly [FB4]

Assembly Properties

General

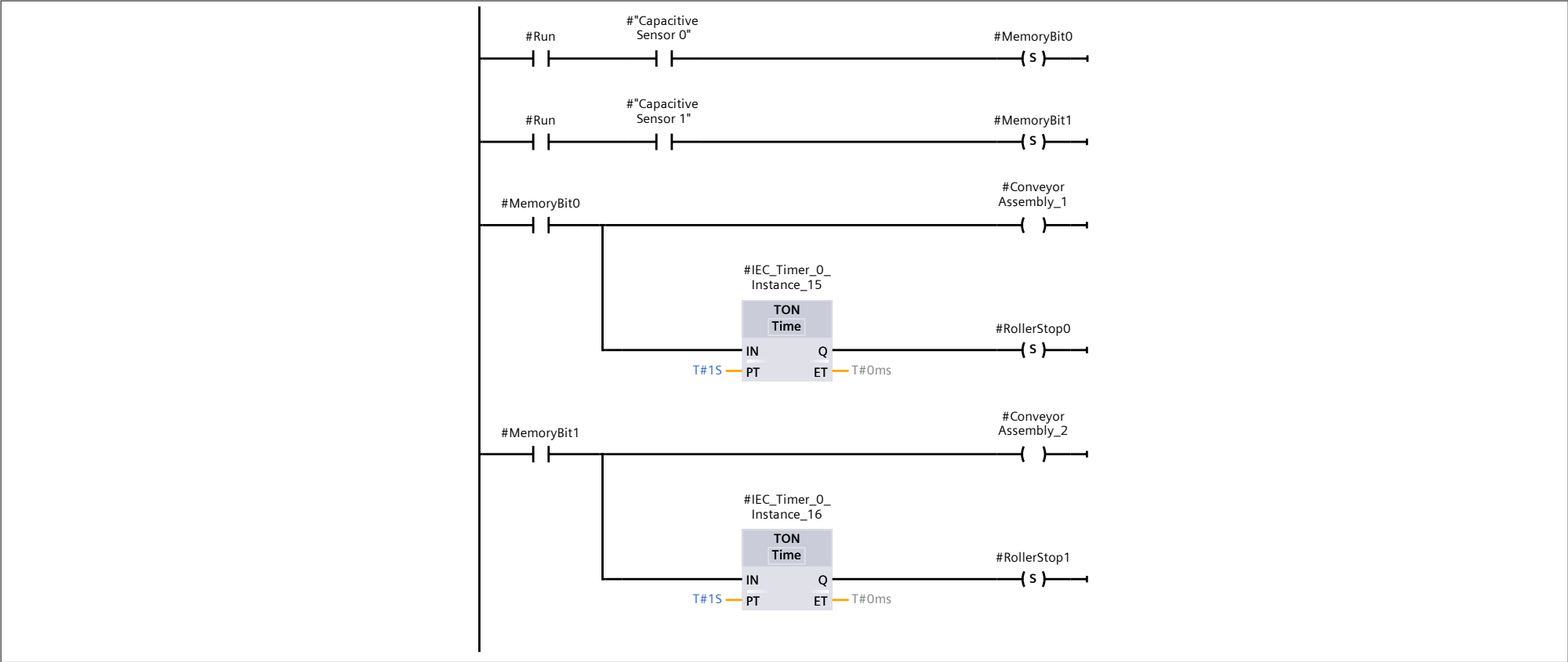
Name	Assembly	Number	4	Type	FB	Language	LAD
Numbering	Automatic						

Information

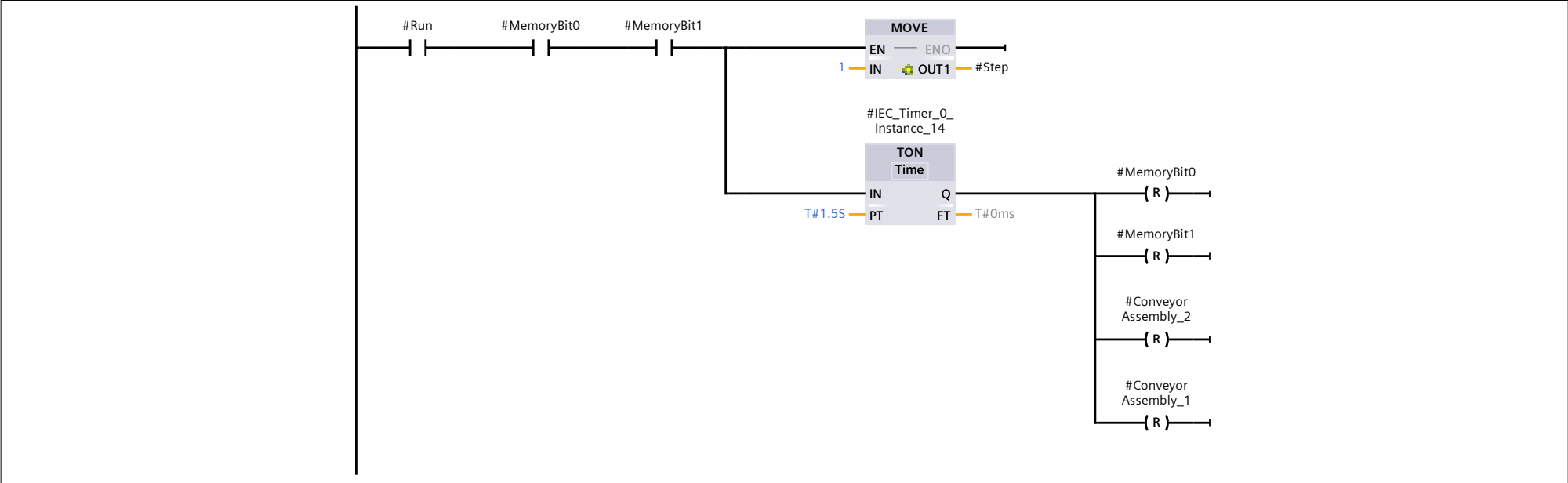
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
Capacitive Sensor 0	Bool	false	Non-retain
Capacitive Sensor 1	Bool	false	Non-retain
RightLimit0	Bool	false	Non-retain
RightLimit1	Bool	false	Non-retain
Two-Axis Pick & Place 0 (Detected)	Bool	false	Non-retain
RightRaiseLimit0	Bool	false	Non-retain
Run	Bool	false	Non-retain
Reset	Bool	false	Non-retain
▼ Output			
MoveRight0	Bool	false	Non-retain
MoveRight1	Bool	false	Non-retain
Two-Axis Pick & Place 0 X	Bool	false	Non-retain
Two-Axis Pick & Place 0 Z	Bool	false	Non-retain
Two-Axis Pick & Place 0 (Grab)	Bool	false	Non-retain
RightRaise0	Bool	false	Non-retain
RollerStop0	Bool	false	Non-retain
RollerStop1	Bool	false	Non-retain
ConveyorAssembly_2	Bool	false	Non-retain
ConveyorAssembly_1	Bool	false	Non-retain
InOut			
▼ Static			
IEC_Timer_0_Instance	TP_TIME		Non-retain
IEC_Timer_0_Instance_1	TP_TIME		Non-retain
IEC_Timer_0_Instance_2	TON_TIME		Non-retain
IEC_Timer_0_Instance_3	TON_TIME		Non-retain
IEC_Timer_0_Instance_4	TON_TIME		Non-retain
MemoryBit1	Bool	false	Non-retain
MemoryBit0	Bool	false	Non-retain
Step	Int	0	Non-retain
IEC_Timer_0_Instance_5	TON_TIME		Non-retain
IEC_Timer_0_Instance_6	TON_TIME		Non-retain
IEC_Timer_0_Instance_7	TON_TIME		Non-retain
IEC_Timer_0_Instance_8	TON_TIME		Non-retain
IEC_Timer_0_Instance_9	TON_TIME		Non-retain
IEC_Timer_0_Instance_10	TON_TIME		Non-retain
IEC_Timer_0_Instance_11	TON_TIME		Non-retain
IEC_Timer_0_Instance_12	TON_TIME		Non-retain
IEC_Timer_0_Instance_13	TON_TIME		Non-retain
IEC_Timer_0_Instance_14	TON_TIME		Non-retain
IEC_Timer_0_Instance_15	TON_TIME		Non-retain
IEC_Timer_0_Instance_16	TON_TIME		Non-retain
IEC_Timer_0_Instance_17	TP_TIME		Non-retain
Temp			
Constant			

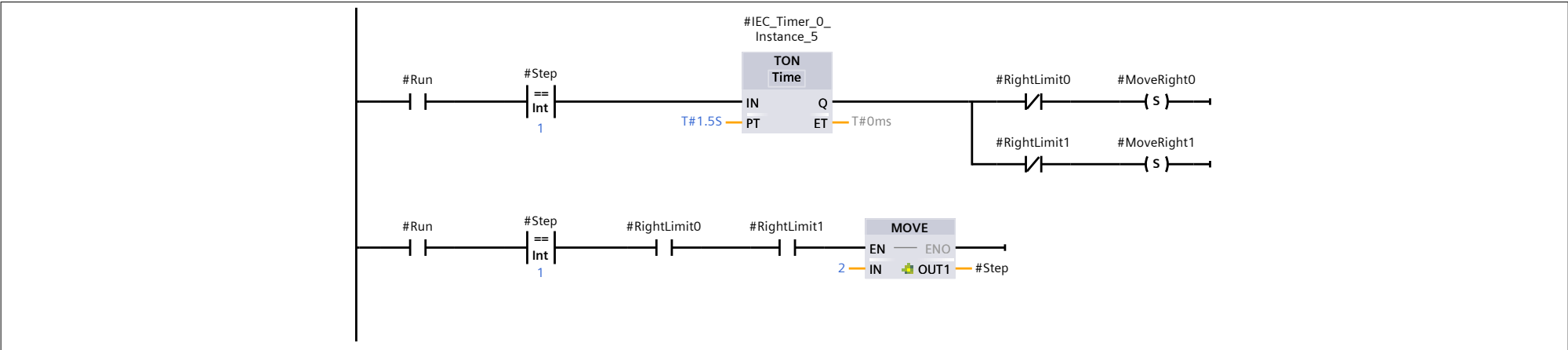
Network 1:



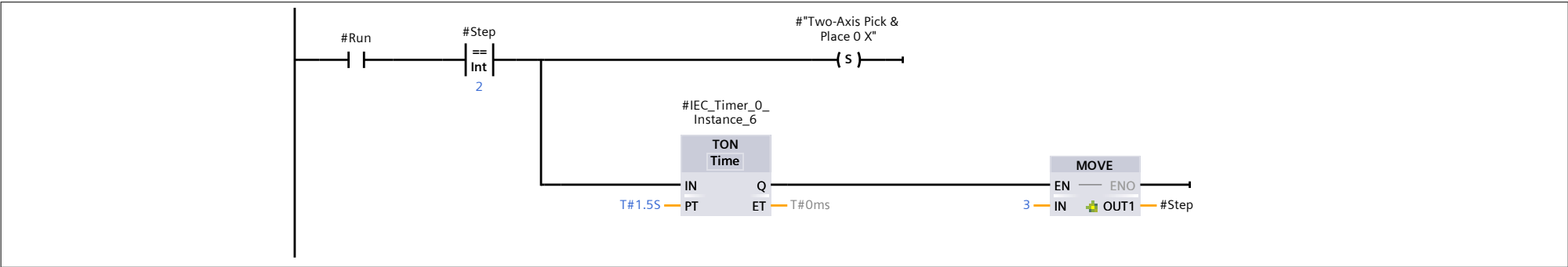
Network 2:



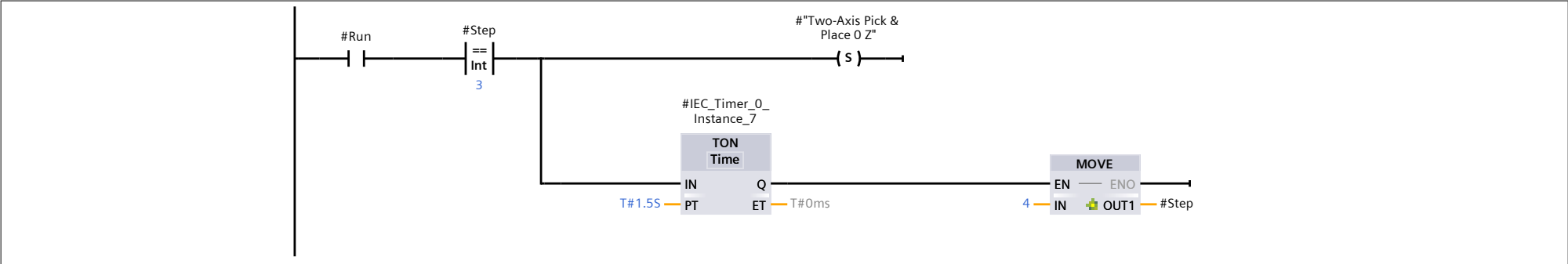
Network 3:



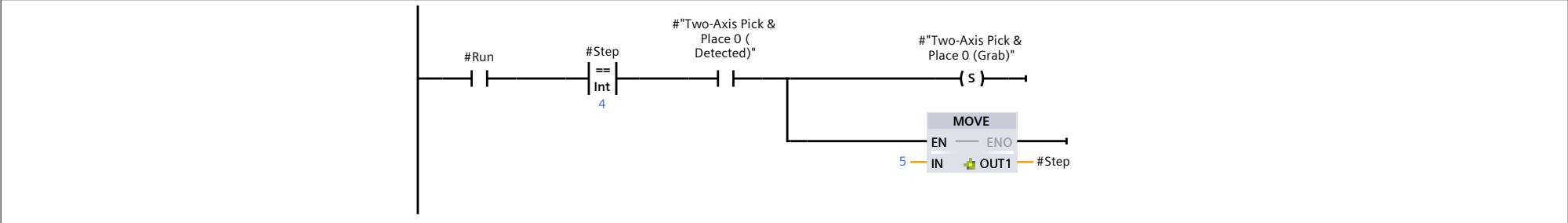
Network 4:



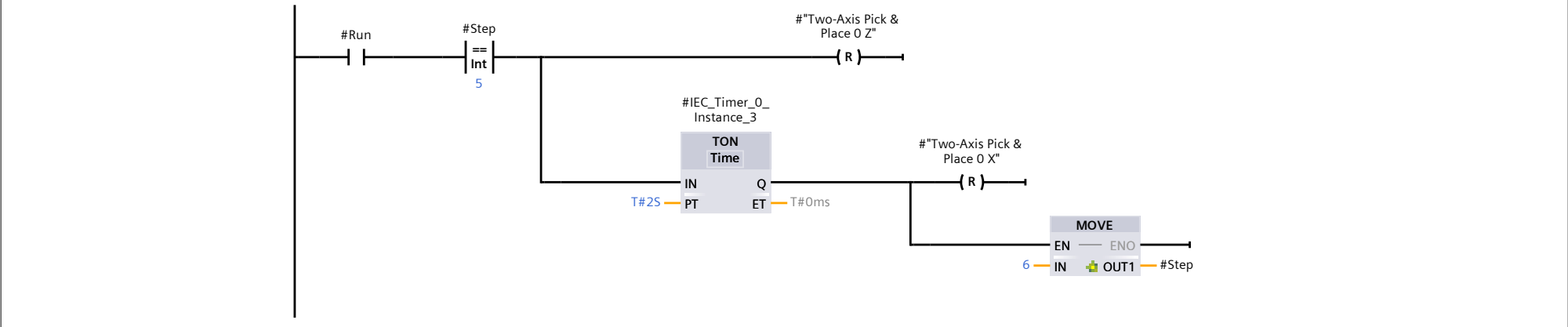
Network 5:



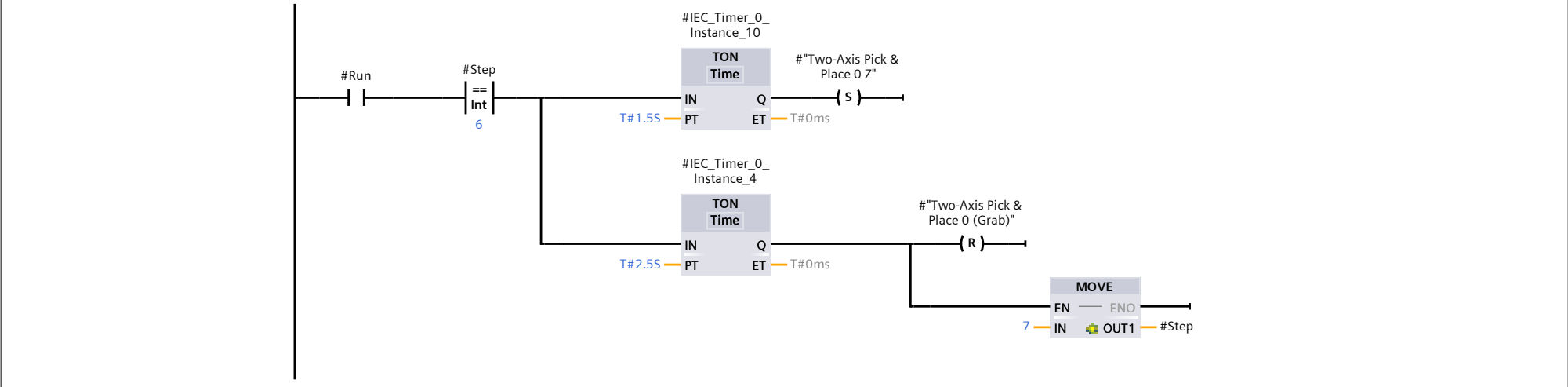
Network 6:



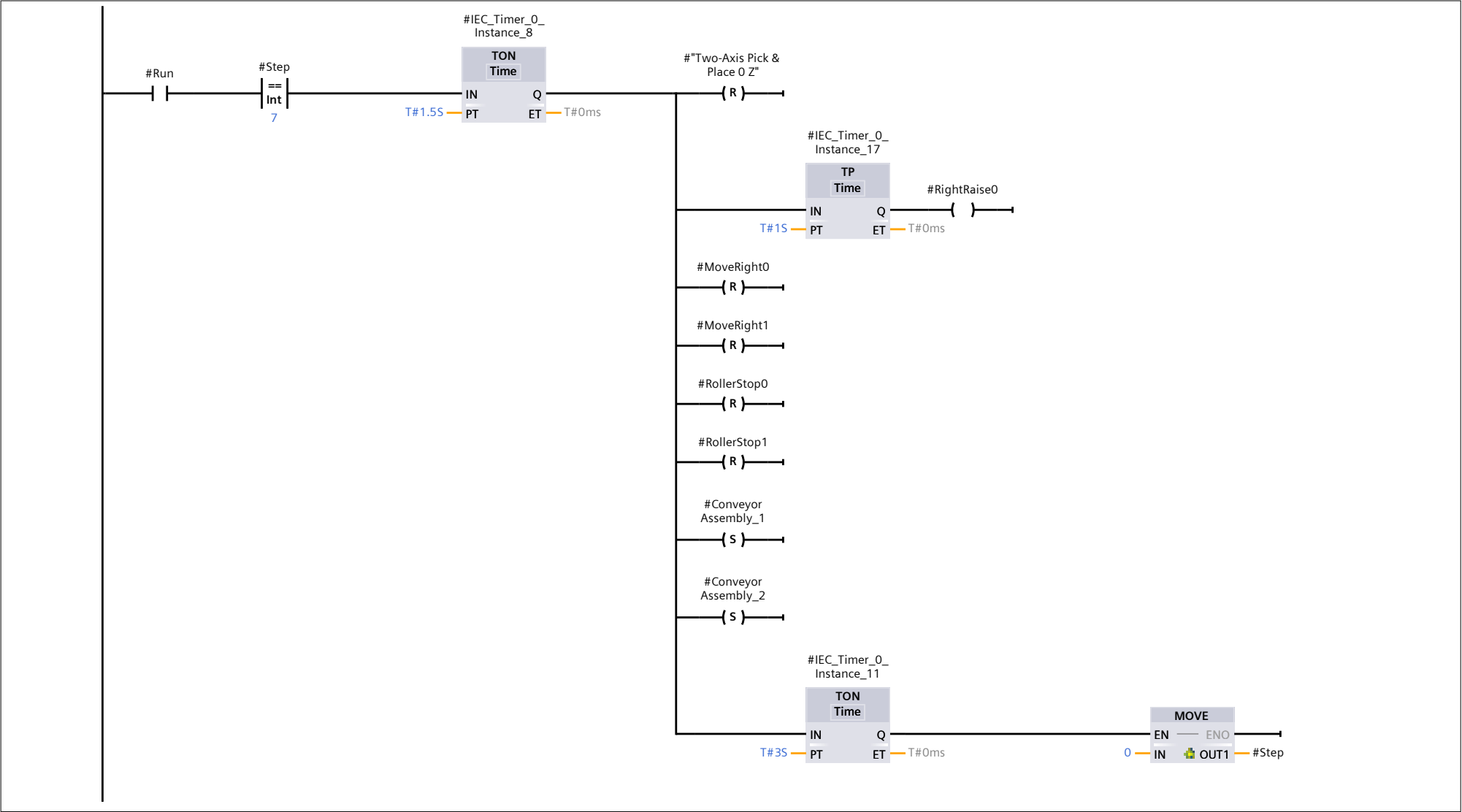
Network 7:



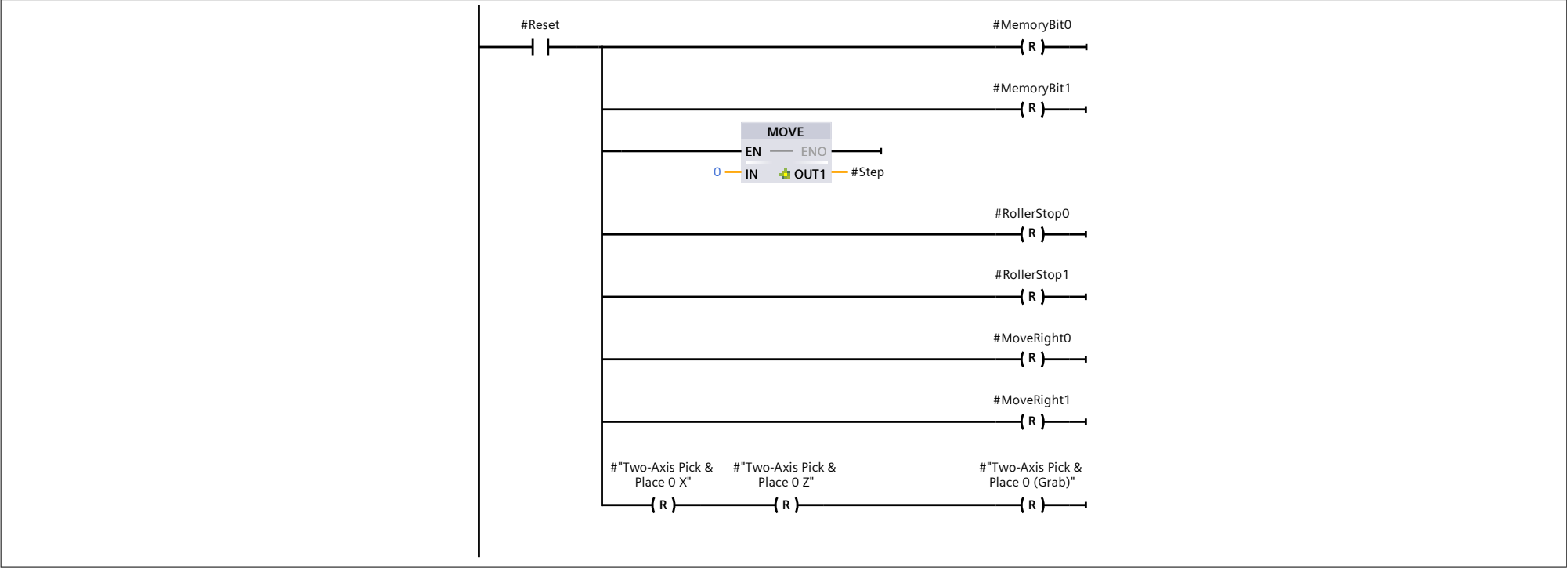
Network 8:



Network 9:



Network 10: Reset



Machine Center Funtion [FB1]

Machine Center Funtion Properties							
General							
Name	Machine Center Funtion	Number	1	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
FCIO	Bool	false	Non-retain
SensorLidsIntrare	Bool	false	Non-retain
SensorLidslesire	Bool	false	Non-retain
Machine_Error	Bool	false	Non-retain
Machine_Busy	Bool	false	Non-retain
PanelStopButton	Bool	false	Non-retain
PanelResetButton	Bool	false	Non-retain
run	Bool	false	Non-retain
▼ Output			
Conveyor_Input	Bool	false	Non-retain
Start_Machine	Bool	false	Non-retain
Counter	DInt	0	Non-retain
Conveyor_Output	Bool	false	Non-retain
Stop_MachineCenter	Bool	false	Non-retain
Reset_MachineCenter	Bool	false	Non-retain
ErrorMachine	Bool	false	Non-retain
InOut			
▼ Static			
MC-0	Bool	false	Non-retain
IEC_Counter_0_Instance	CTU_INT		Non-retain
IEC_Timer_0_Instance	TON_TIME		Non-retain
IEC_Timer_0_Instance_1	TON_TIME		Non-retain
Tag_3	Bool	false	Non-retain
orice	Bool	false	Non-retain
IEC_Counter_0_Instance_1	CTU_INT		Non-retain
IEC_Timer_0_Instance_2	TP_TIME		Non-retain
▼ Temp			
Tag_5	Bool		
orice1	Bool		
Constant			

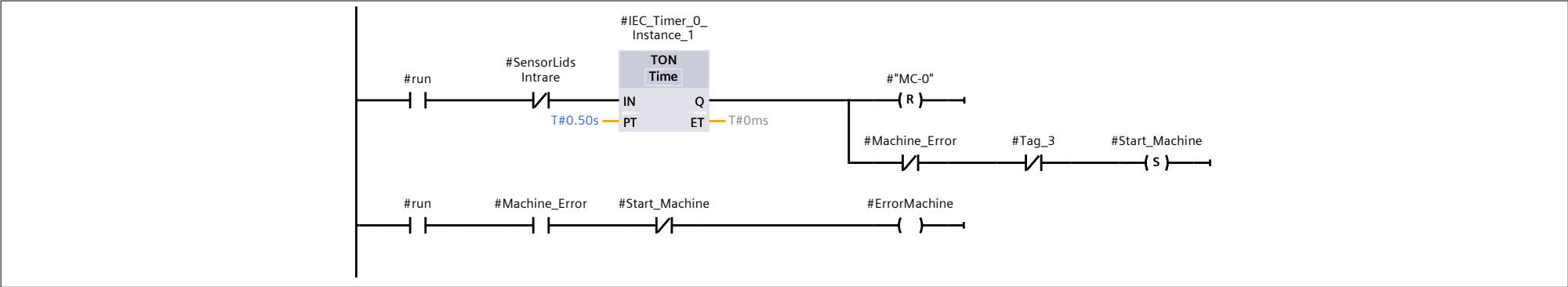
Network 1:



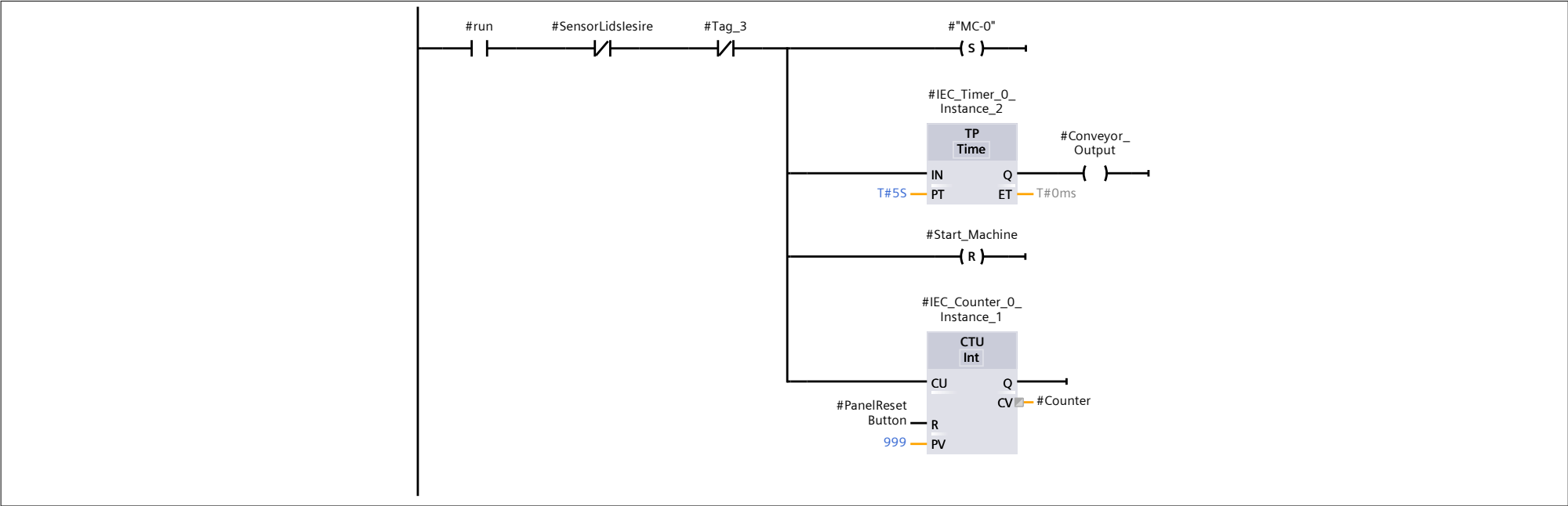
Network 2:



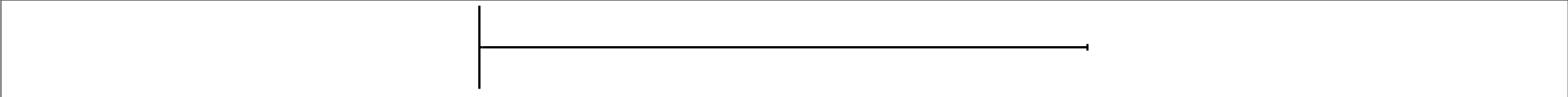
Network 3:



Network 4:



Network 5:



PaletLogic [FB5]

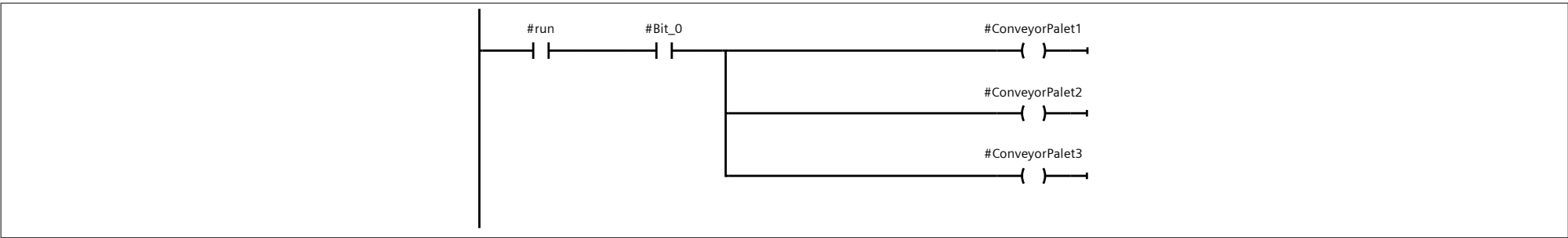
PaletLogic Properties							
General							
Name	PaletLogic	Number	5	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
DiffuseSensorPalet	Bool	false	Non-retain
DiffuseSensorPalet1	Bool	false	Non-retain
LightEmitter	DInt	0	Non-retain
DiffuserSensorPickUP	Bool	false	Non-retain
XPosition	Real	0.0	Non-retain
YPosition	Real	0.0	Non-retain
BoxDetected	Bool	false	Non-retain
ZPosition	Real	0.0	Non-retain
Light Array Emitter 0 (Value)	DInt	0	Non-retain
run	Bool	false	Non-retain
Reset	Bool	false	Non-retain
▼ Output			
ConveyorPalet1	Bool	false	Non-retain
ConveyorPalet2	Bool	false	Non-retain
Emitter	Bool	false	Non-retain
PositionerLeftClamp	Bool	false	Non-retain
YSetPoint	Real	0.0	Non-retain
XSetPoint	Real	0.0	Non-retain
Pick&Place Grab	Bool	false	Non-retain
ZSetPoint	Real	0.0	Non-retain
ConveyorPalet3	Bool	false	Non-retain
ResetButton	Bool	false	Non-retain
InOut			
▼ Static			
NrPiesei	Int	0	Non-retain
IEC_Timer_0_Instance	TP_TIME		Non-retain
IEC_Timer_0_Instance_1	TON_TIME		Non-retain
IEC_Timer_0_Instance_2	TON_TIME		Non-retain
Step	Int	0	Non-retain
IEC_Counter_0_Instance	CTU_INT		Retain
R_TRIG_Instance	R_TRIG		
R_TRIG_Instance_1	R_TRIG		
Bit_0	Bool	false	Non-retain
IEC_Timer_0_Instance_3	TP_TIME		Non-retain
IEC_Counter_0_Instance_1	CTD_INT		Retain
R_TRIG_Instance_2	R_TRIG		
orice2	Bool	false	Non-retain
▼ Temp			
Orice	Bool		
orice1	Bool		
Constant			

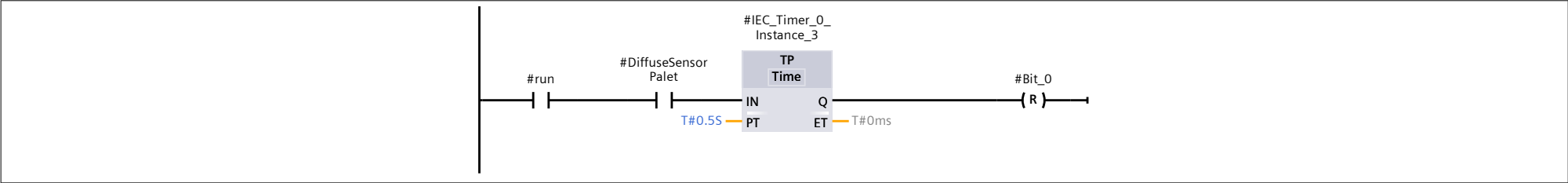
Network 1:



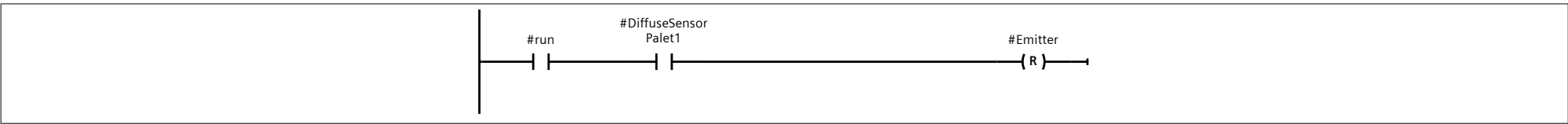
Network 2:



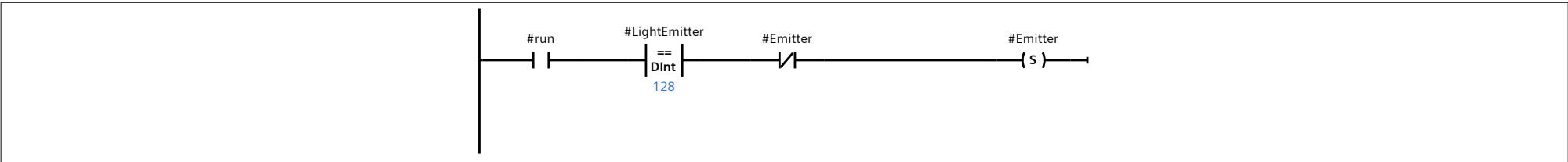
Network 3:



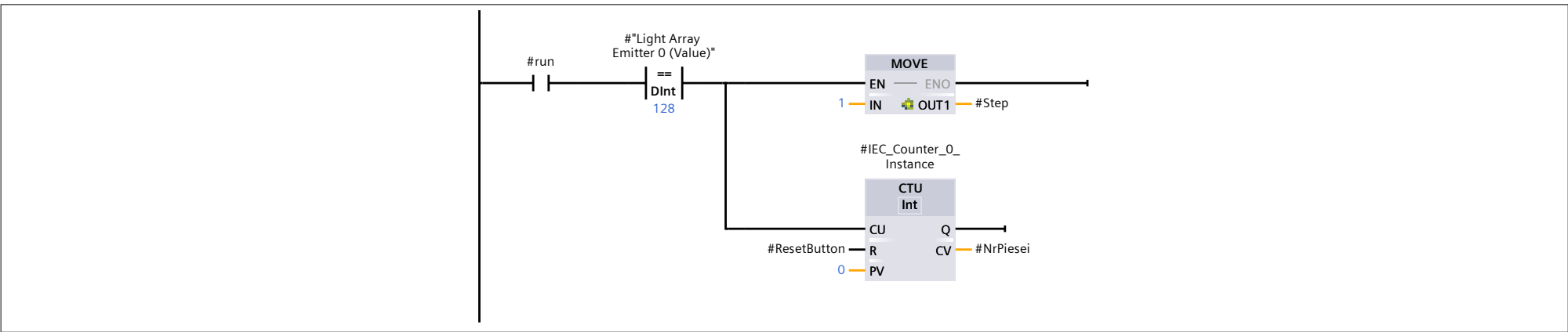
Network 4:



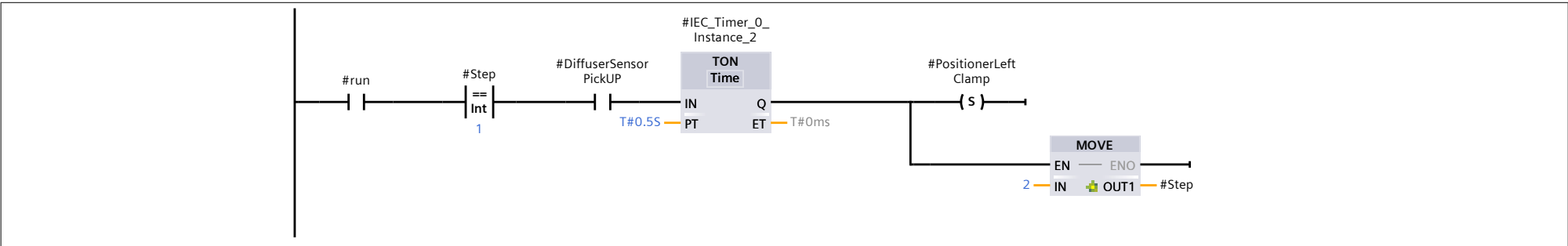
Network 5:



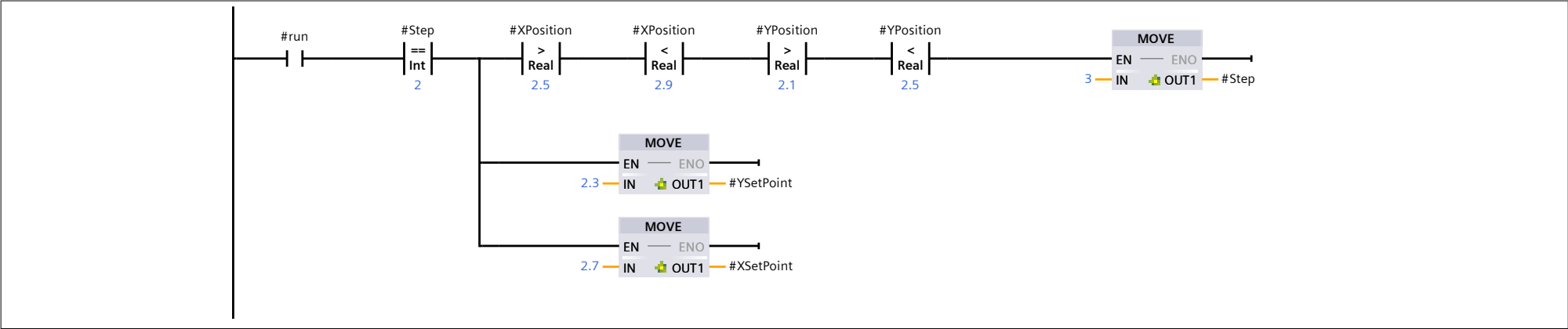
Network 6:



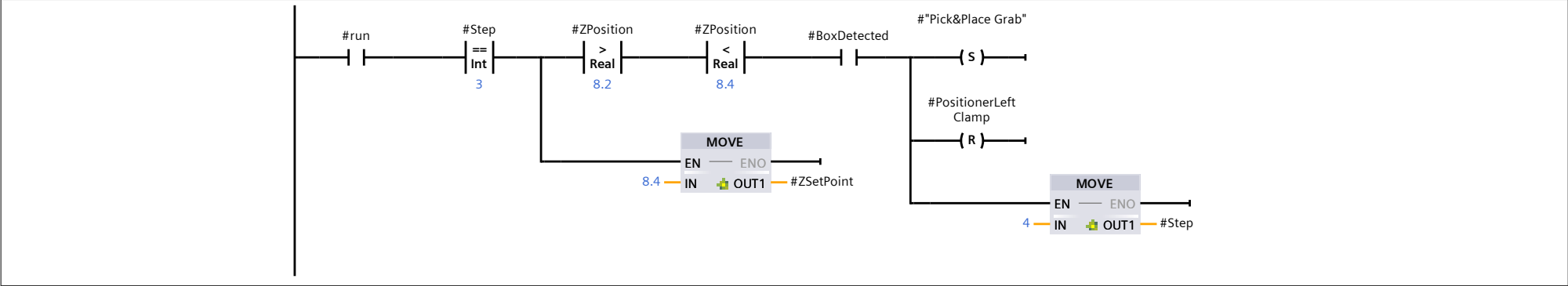
Network 7:



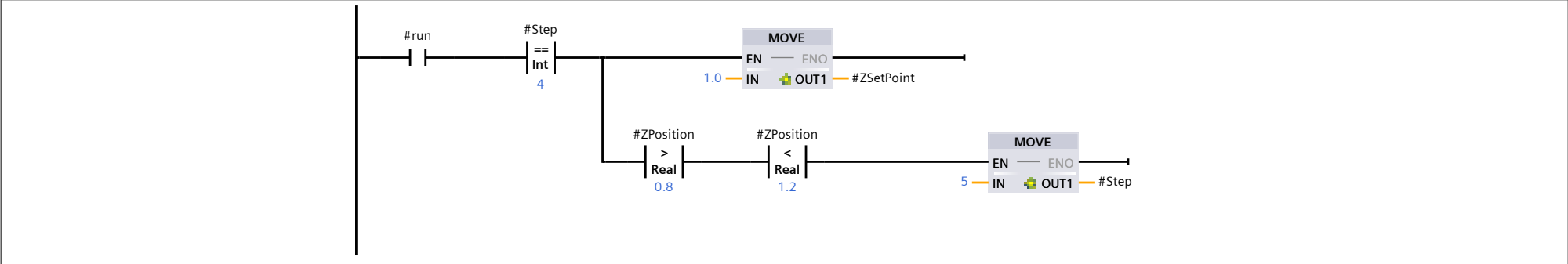
Network 8:



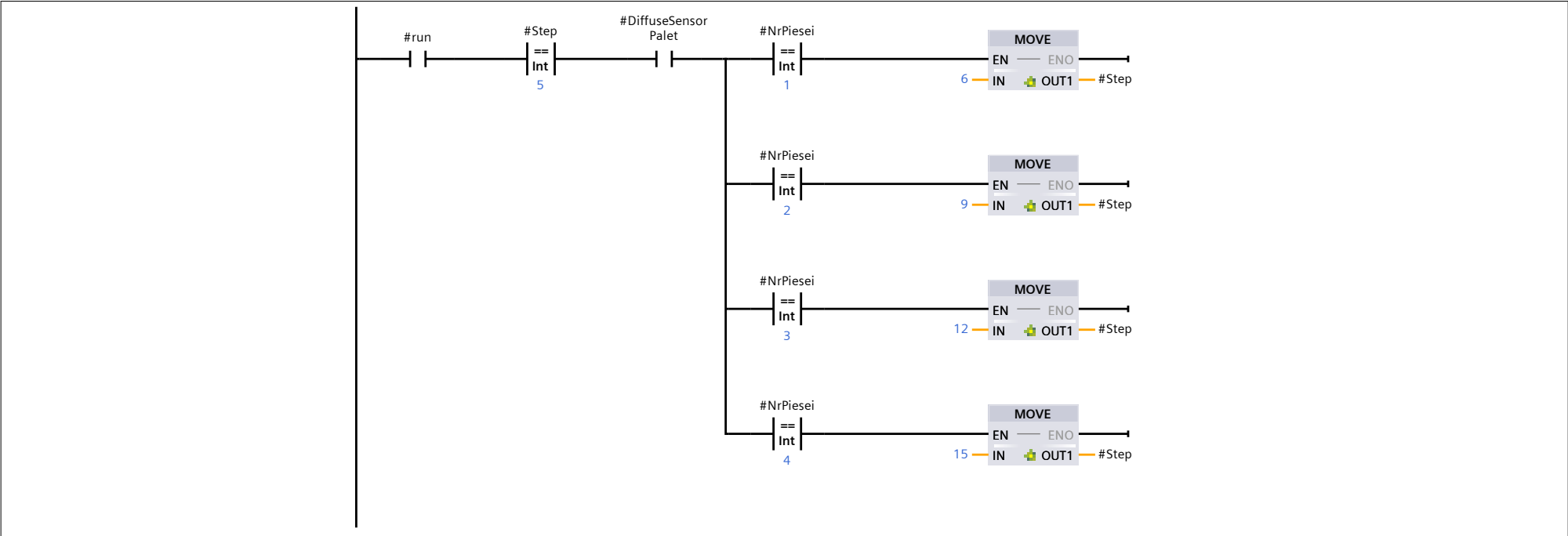
Network 9:



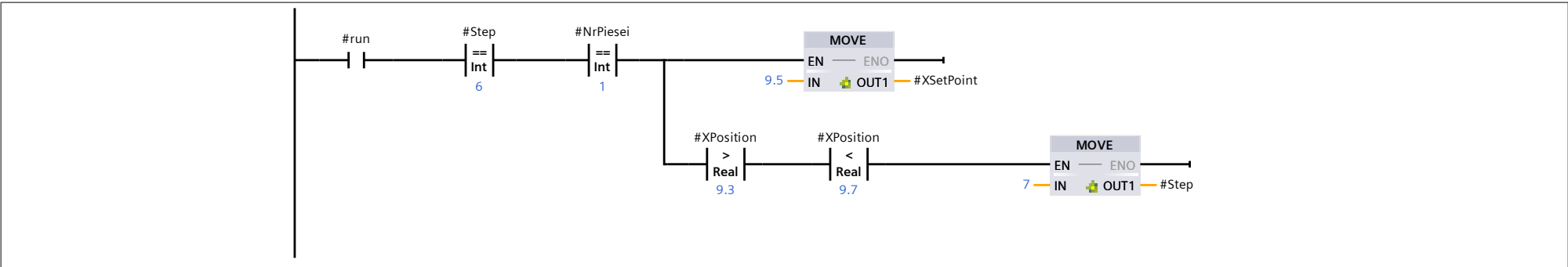
Network 10:



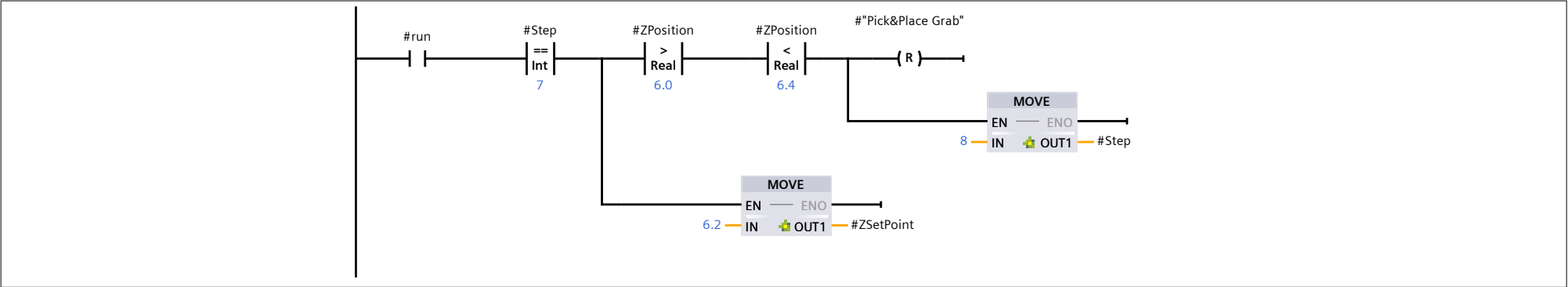
Network 11:



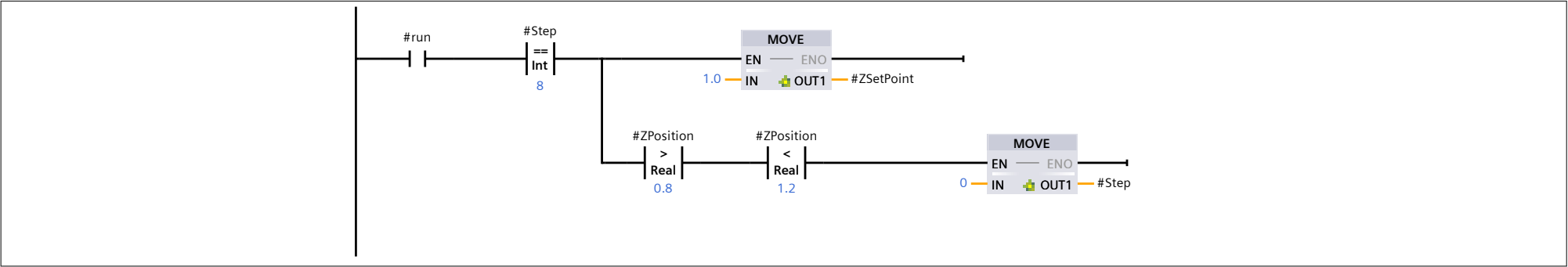
Network 12: Piesa Numarul 1



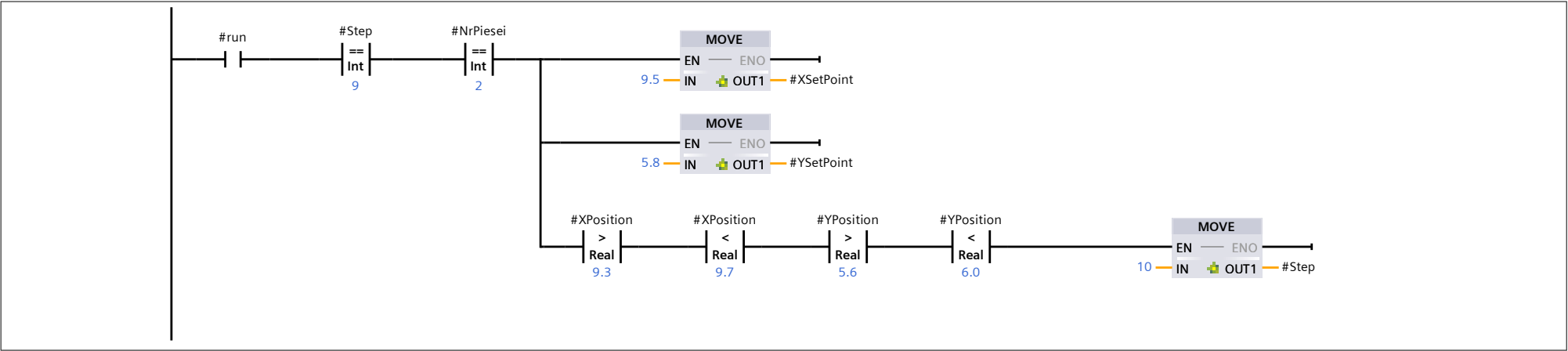
Network 13:



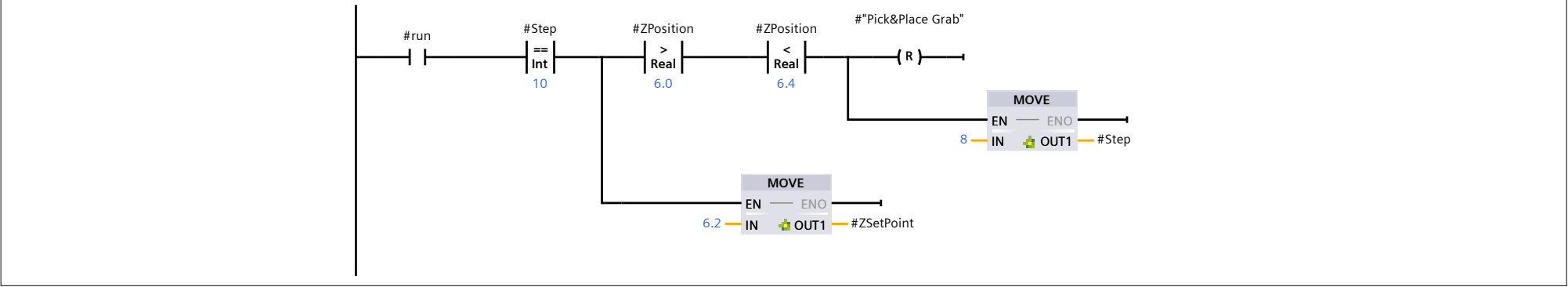
Network 14:



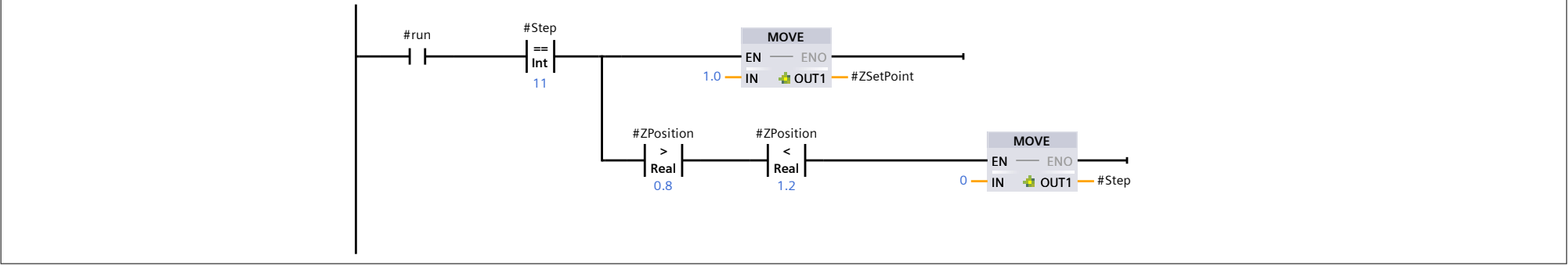
Network 15: Piesa Numarul 2



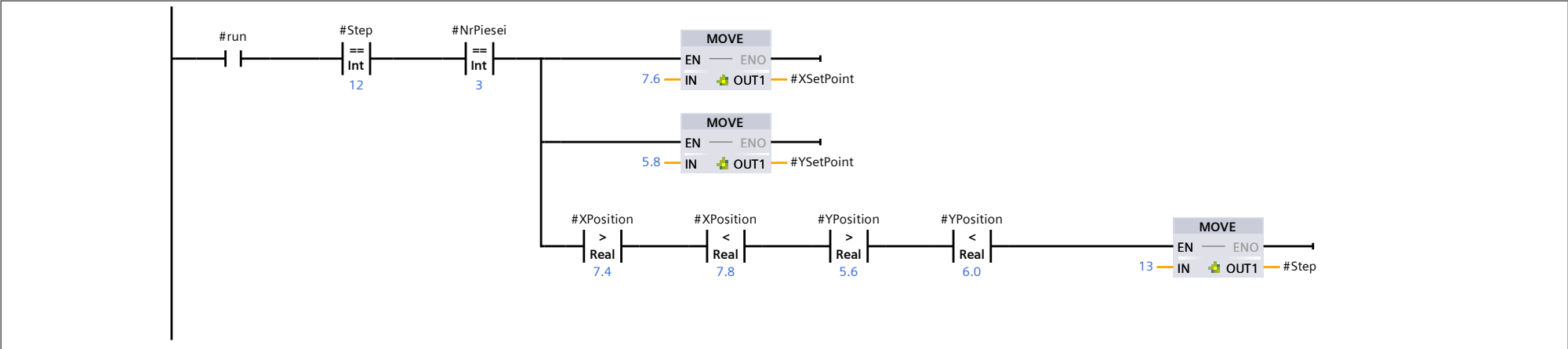
Network 16:



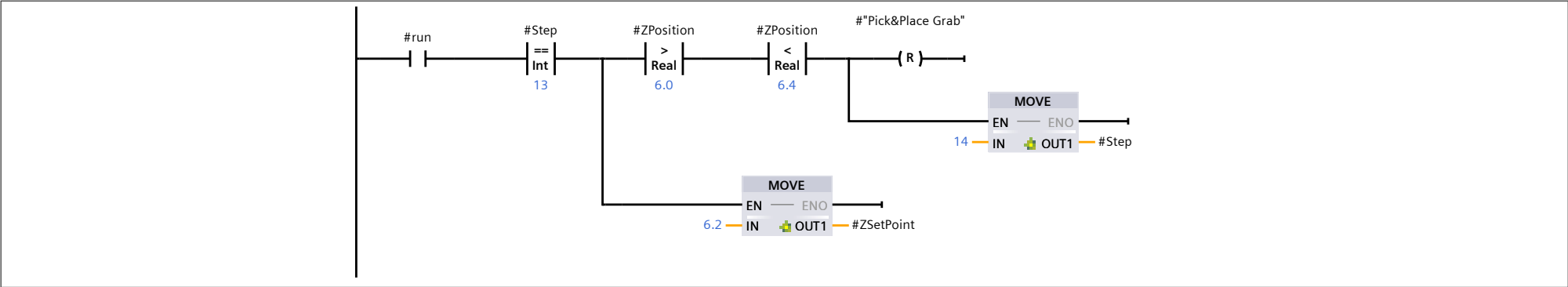
Network 17:



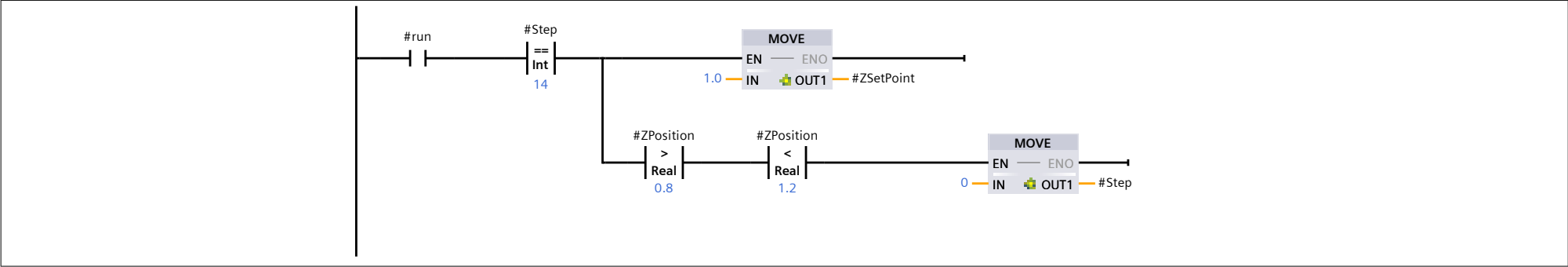
Network 18: Piesa Numarul 3



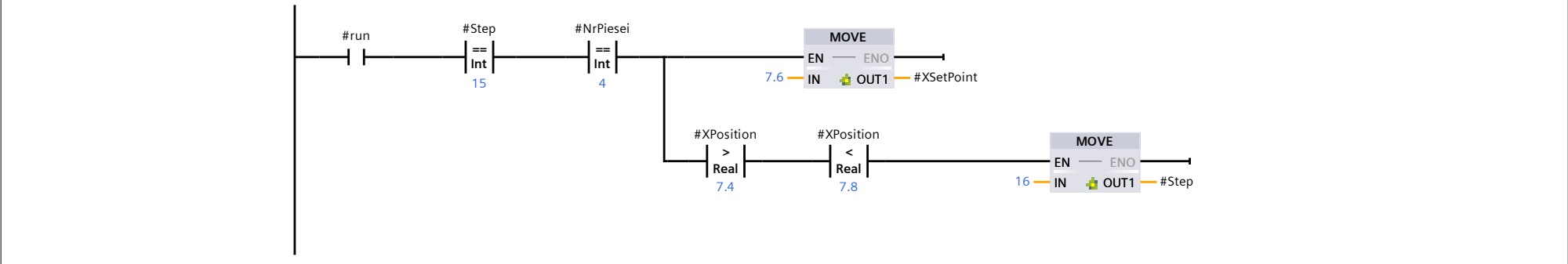
Network 19:



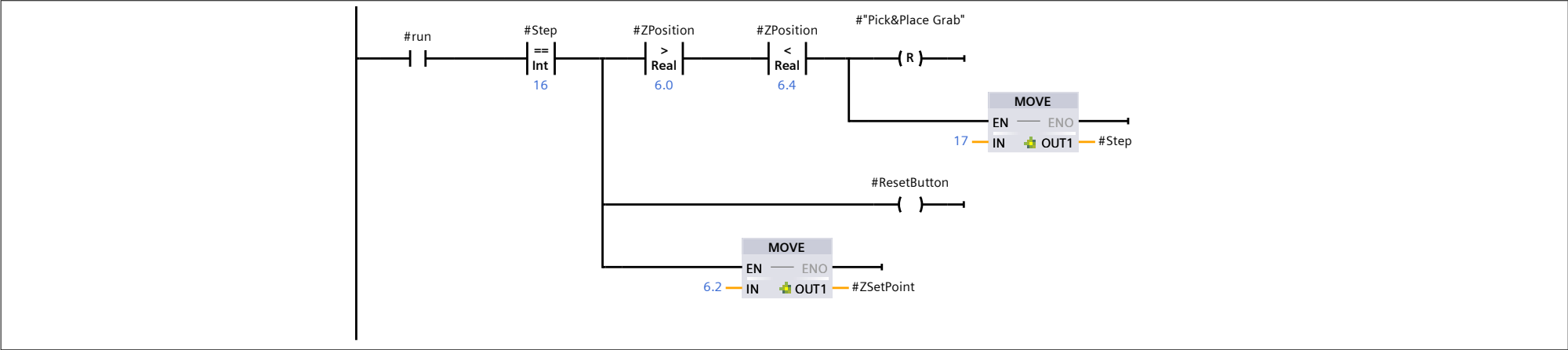
Network 20:



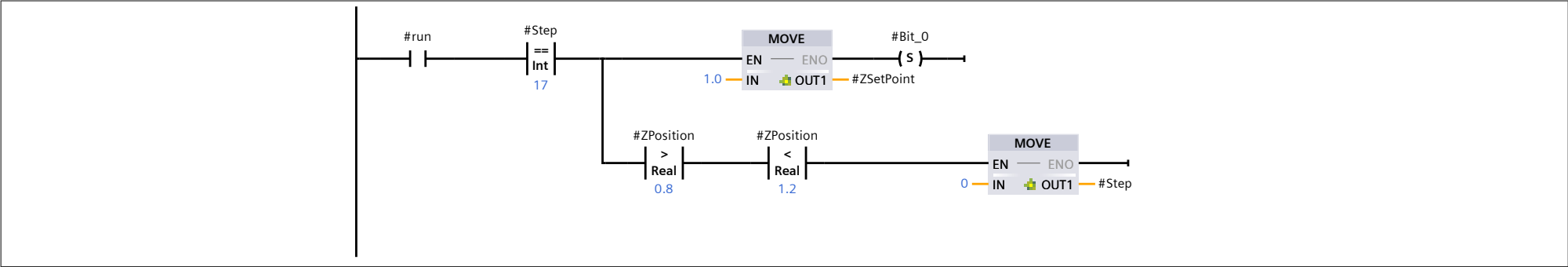
Network 21: Pisea numarul 4



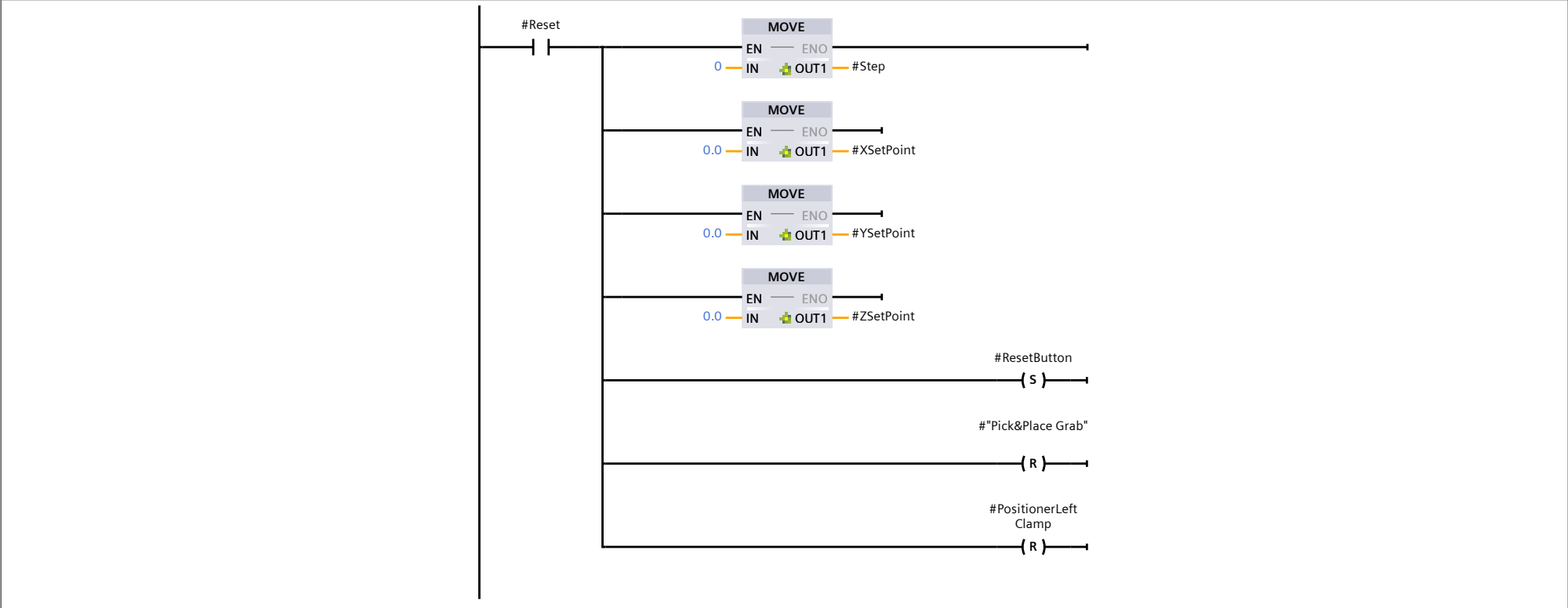
Network 22:



Network 23:



Network 24: Reset

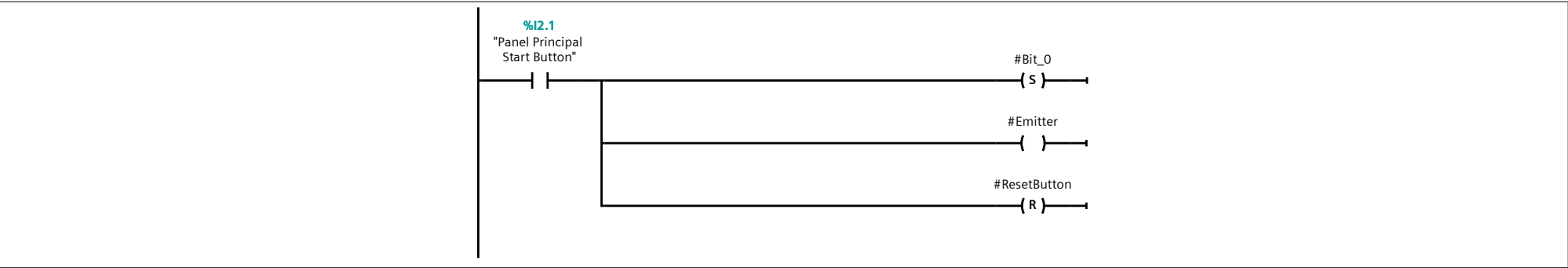


PaletLogic_1 [FB6]

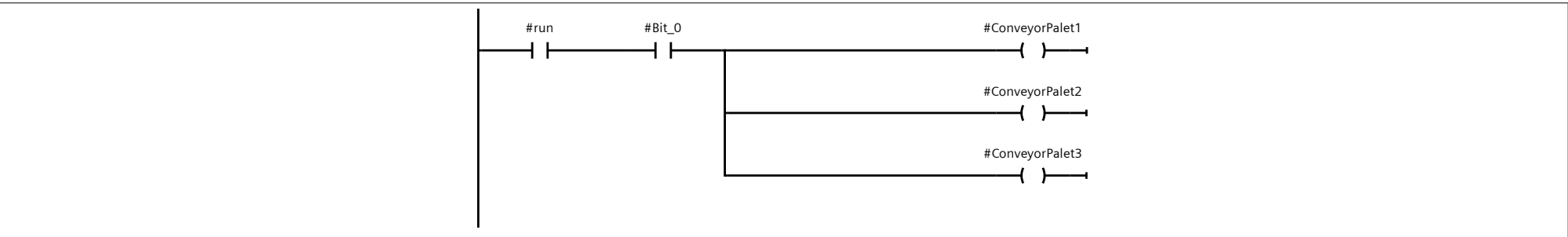
PaletLogic_1 Properties							
General							
Name	PaletLogic_1	Number	6	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
DiffuseSensorPalet	Bool	false	Non-retain
DiffuseSensorPalet1	Bool	false	Non-retain
LightEmitter	DInt	0	Non-retain
DiffuserSensorPickUP	Bool	false	Non-retain
XPosition	Real	0.0	Non-retain
YPosition	Real	0.0	Non-retain
BoxDetected	Bool	false	Non-retain
ZPosition	Real	0.0	Non-retain
Light Array Emitter 0 (Value)	DInt	0	Non-retain
Reset	Bool	false	Non-retain
run	Bool	false	Non-retain
▼ Output			
ConveyorPalet1	Bool	false	Non-retain
ConveyorPalet2	Bool	false	Non-retain
Emitter	Bool	false	Non-retain
PositionerLeftClamp	Bool	false	Non-retain
YSetPoint	Real	0.0	Non-retain
XSetPoint	Real	0.0	Non-retain
Pick&Place Grab	Bool	false	Non-retain
ZSetPoint	Real	0.0	Non-retain
ConveyorPalet3	Bool	false	Non-retain
ResetButton	Bool	false	Non-retain
InOut			
▼ Static			
IEC_Timer_0_Instance	TP_TIME		Non-retain
IEC_Timer_0_Instance_1	TON_TIME		Non-retain
IEC_Timer_0_Instance_2	TON_TIME		Non-retain
Step	Int	0	Non-retain
IEC_Counter_0_Instance	CTU_INT		Retain
NrPiesei	Int	0	Non-retain
R_TRIG_Instance	R_TRIG		
R_TRIG_Instance_1	R_TRIG		
Bit_0	Bool	false	Non-retain
IEC_Timer_0_Instance_3	TP_TIME		Non-retain
▼ Temp			
Orice	Bool		
Constant			

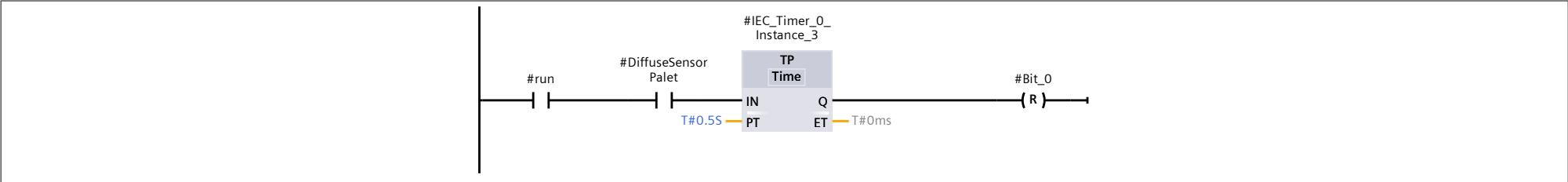
Network 1:



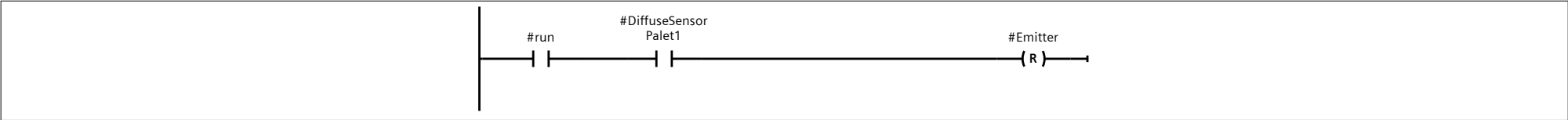
Network 2:



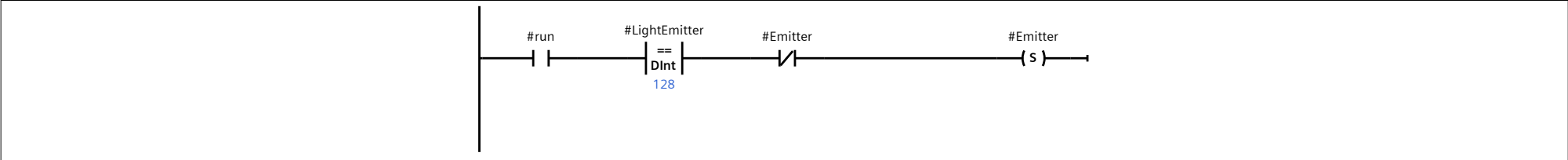
Network 3:



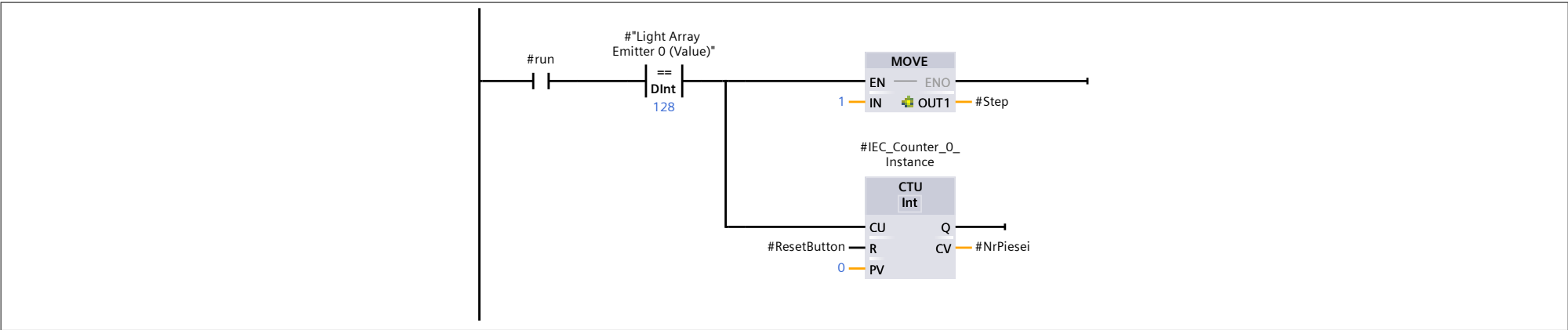
Network 4:



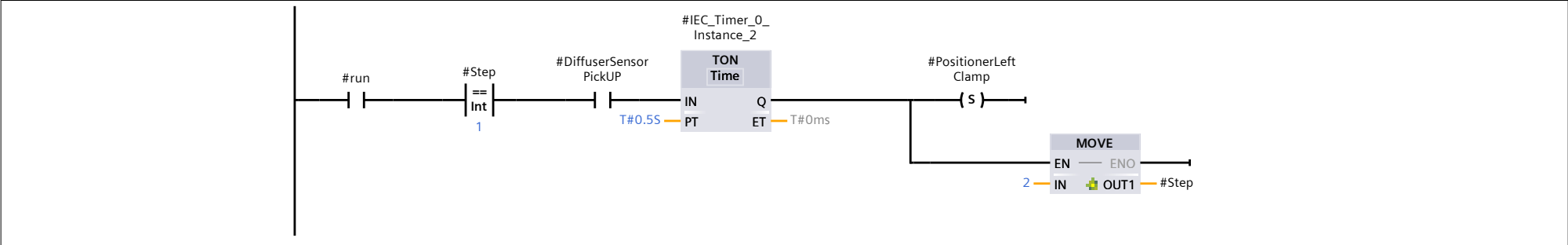
Network 5:



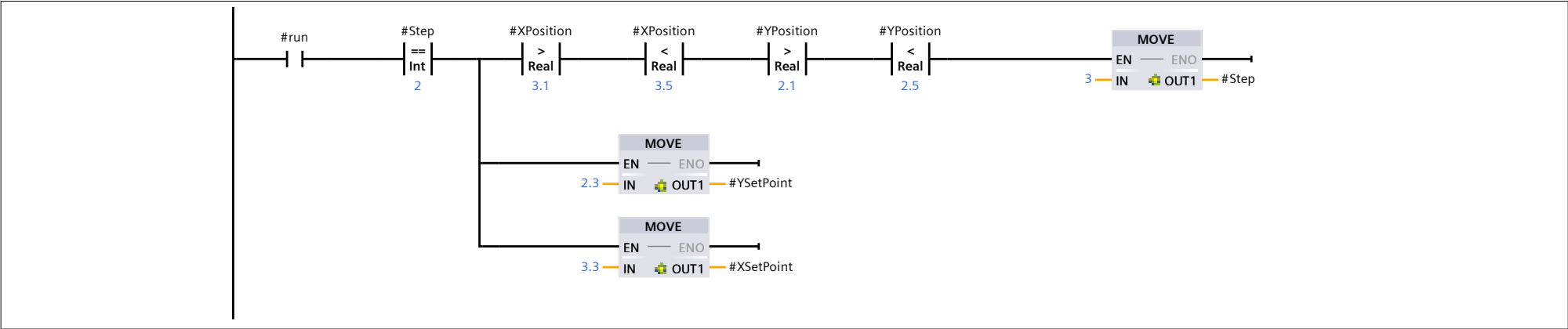
Network 6:



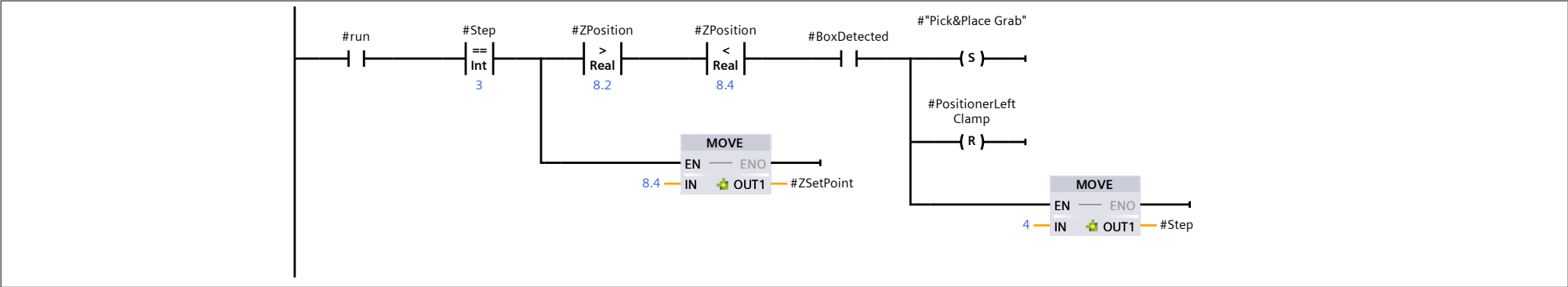
Network 7:



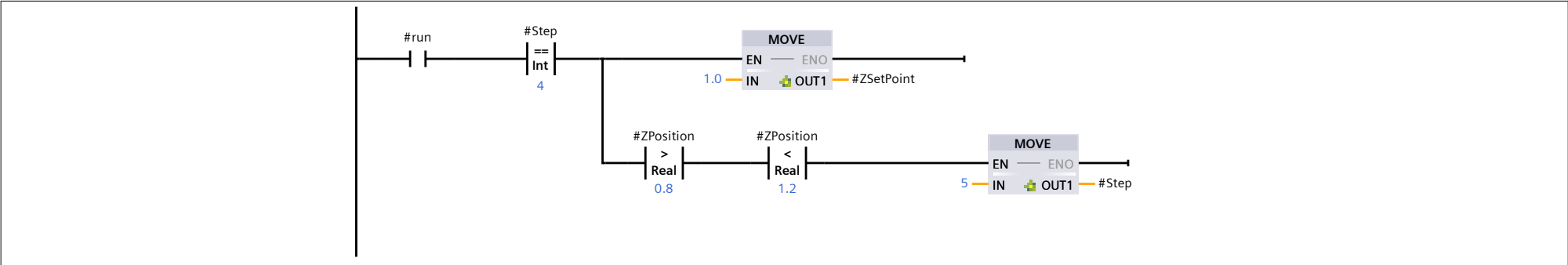
Network 8:



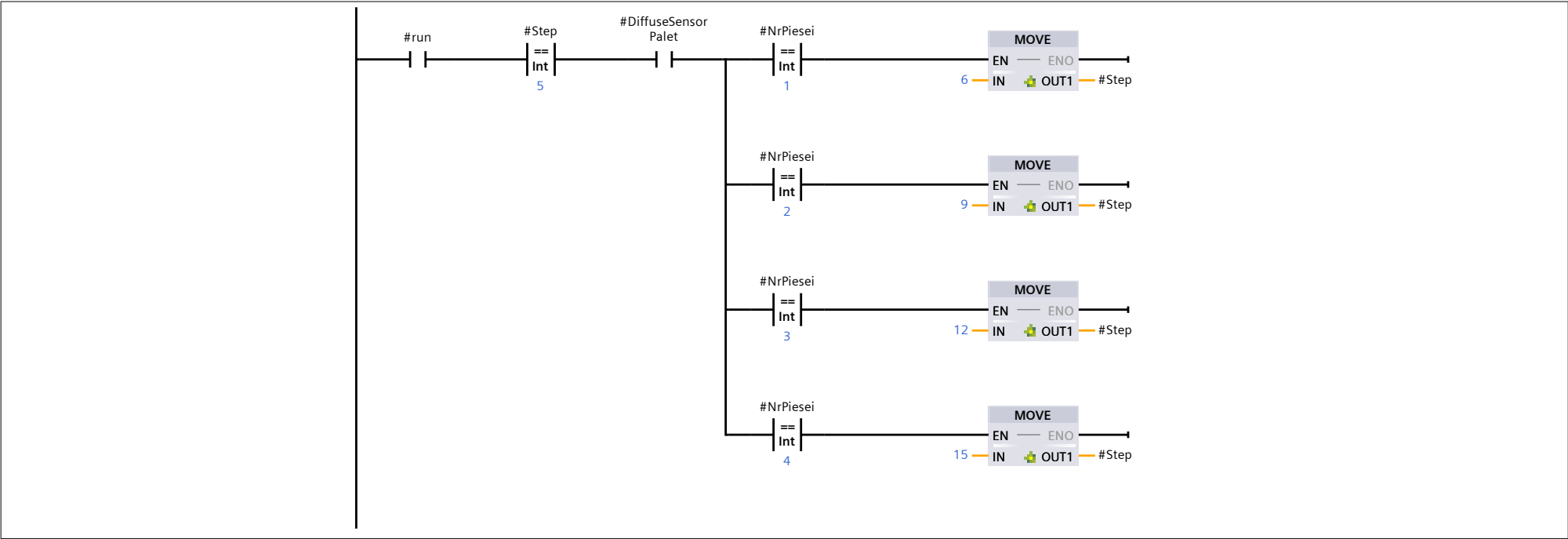
Network 9:



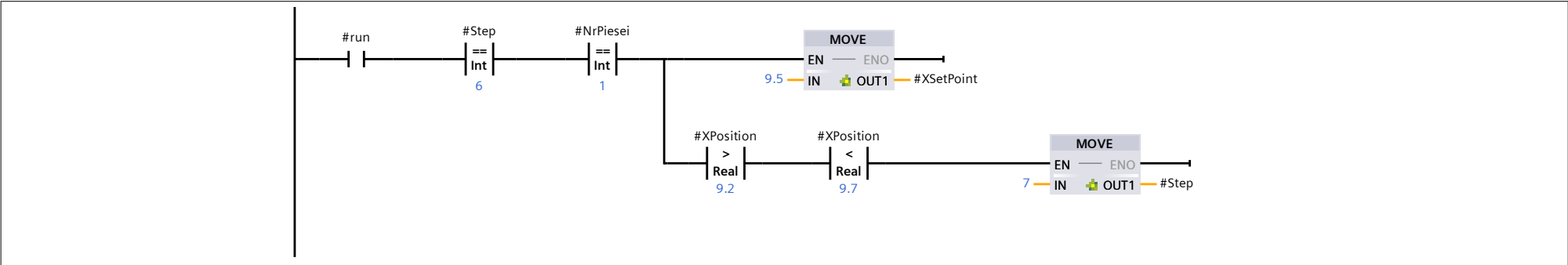
Network 10:



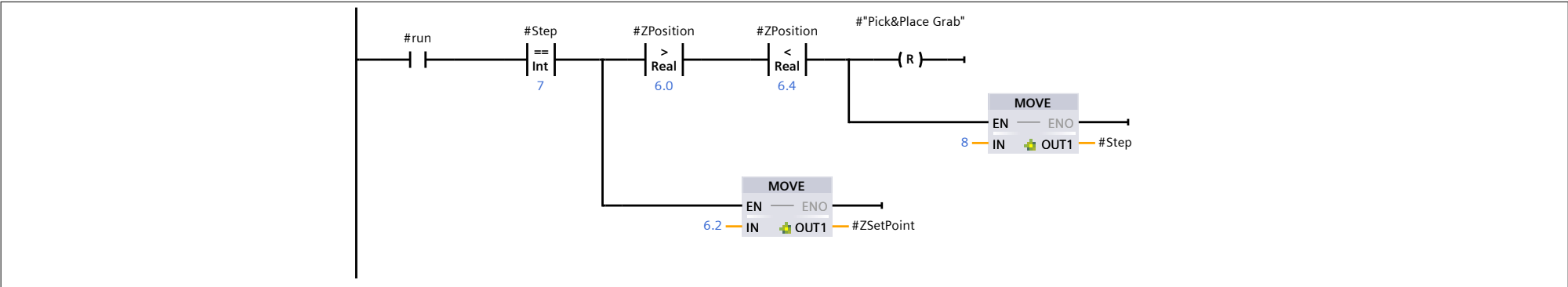
Network 11:



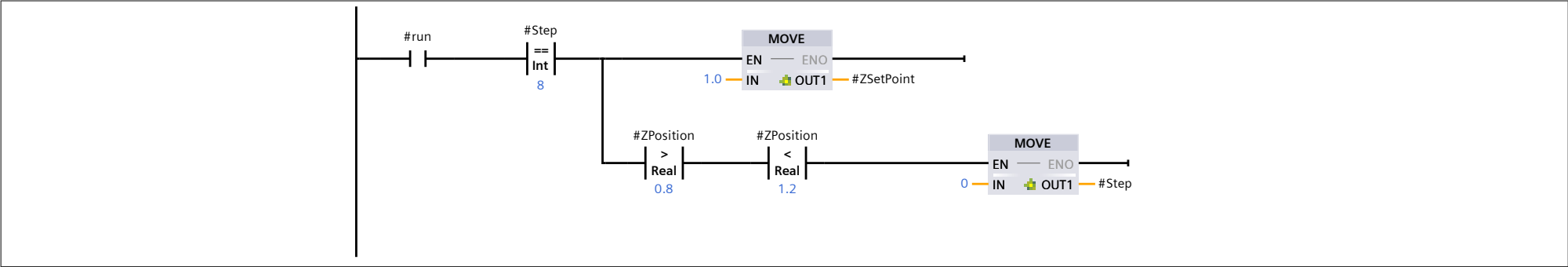
Network 12: Piesa Numarul 1



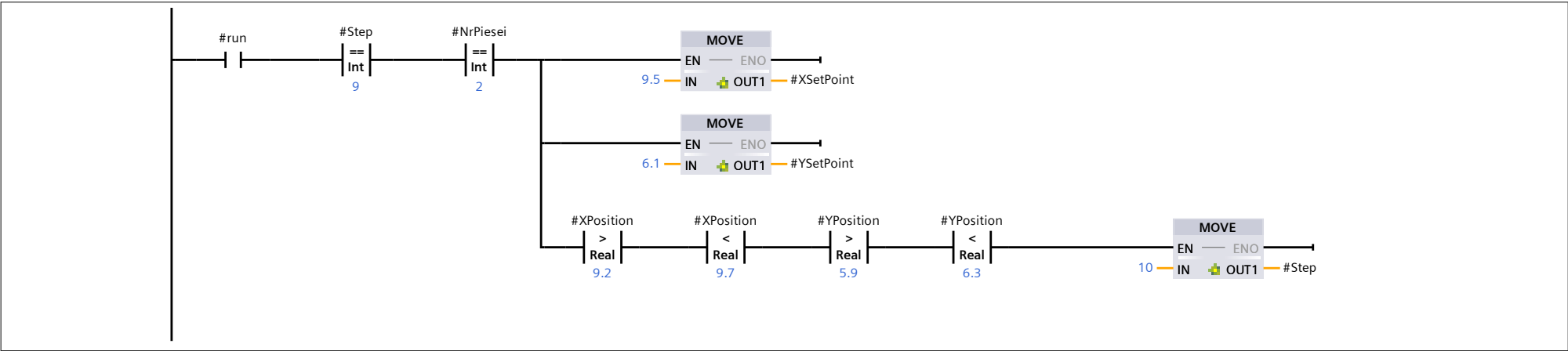
Network 13:



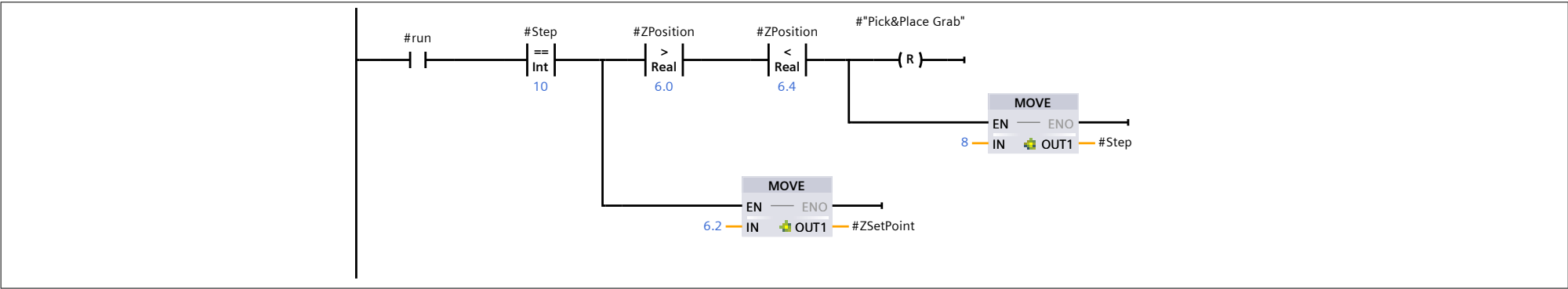
Network 14:



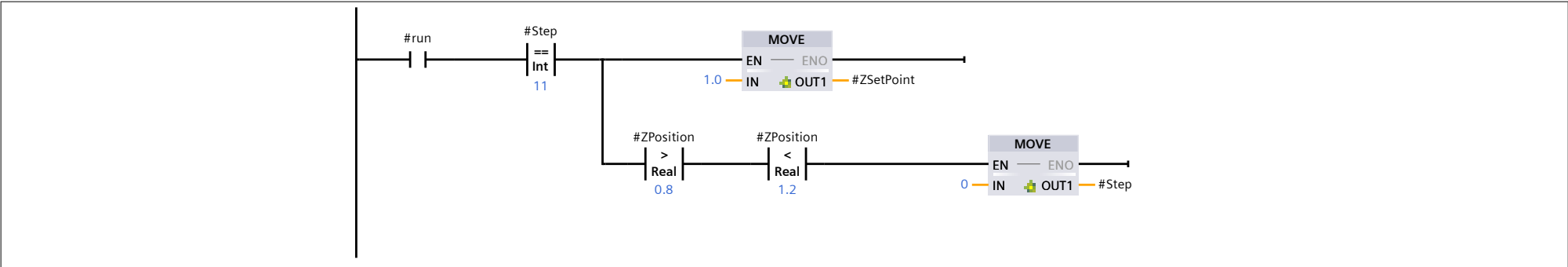
Network 15: Piesa Numarul 2



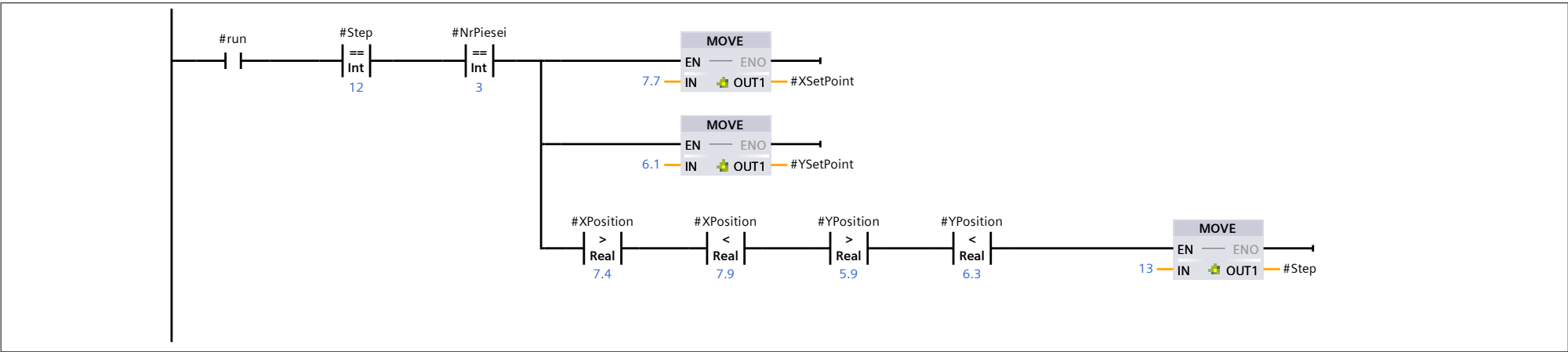
Network 16:



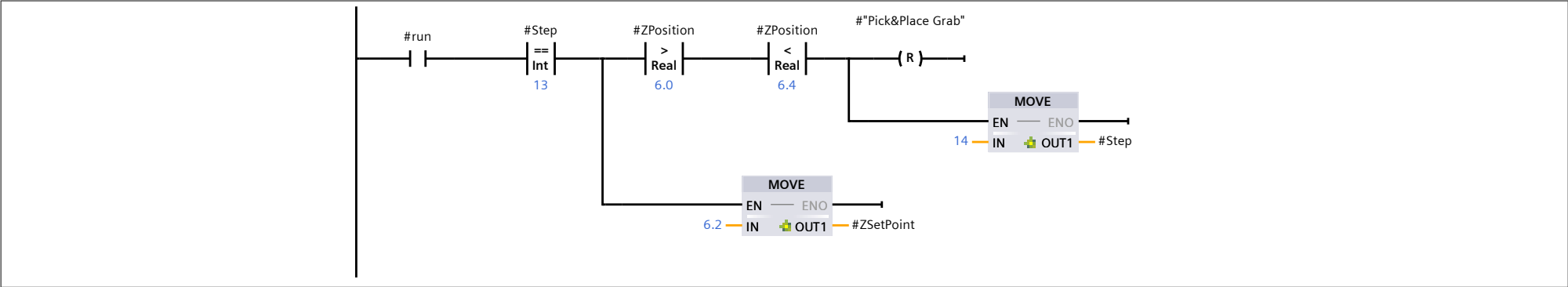
Network 17:



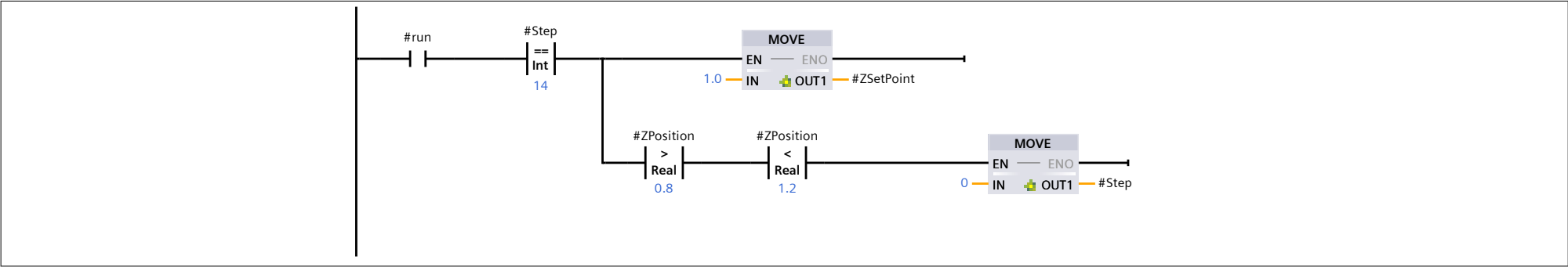
Network 18: Piesa Numarul 3



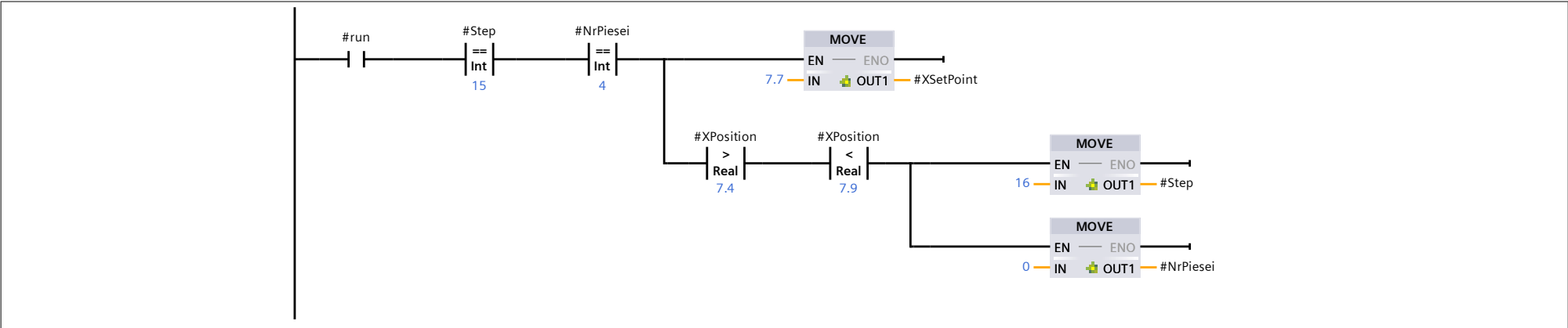
Network 19:



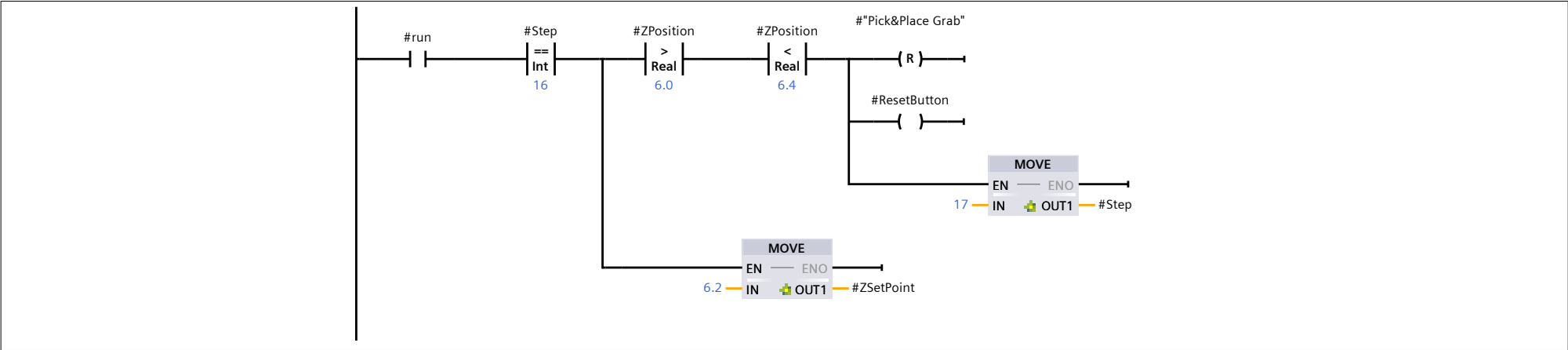
Network 20:



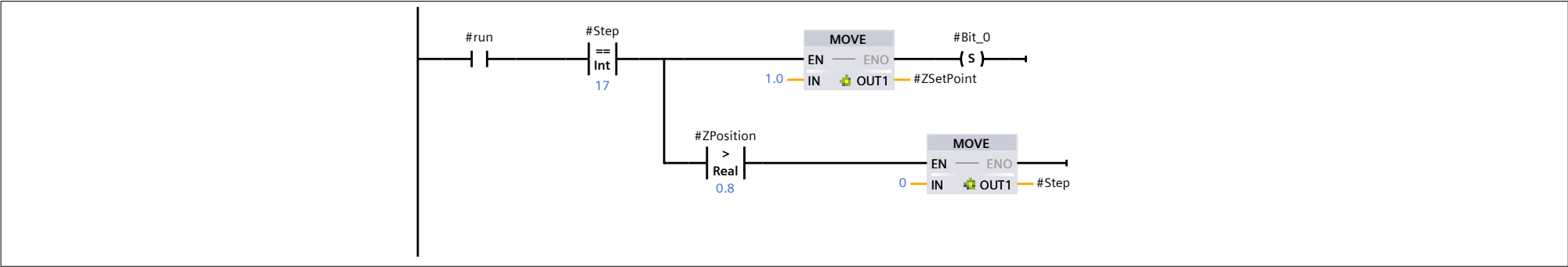
Network 21: Pisea numarul 4



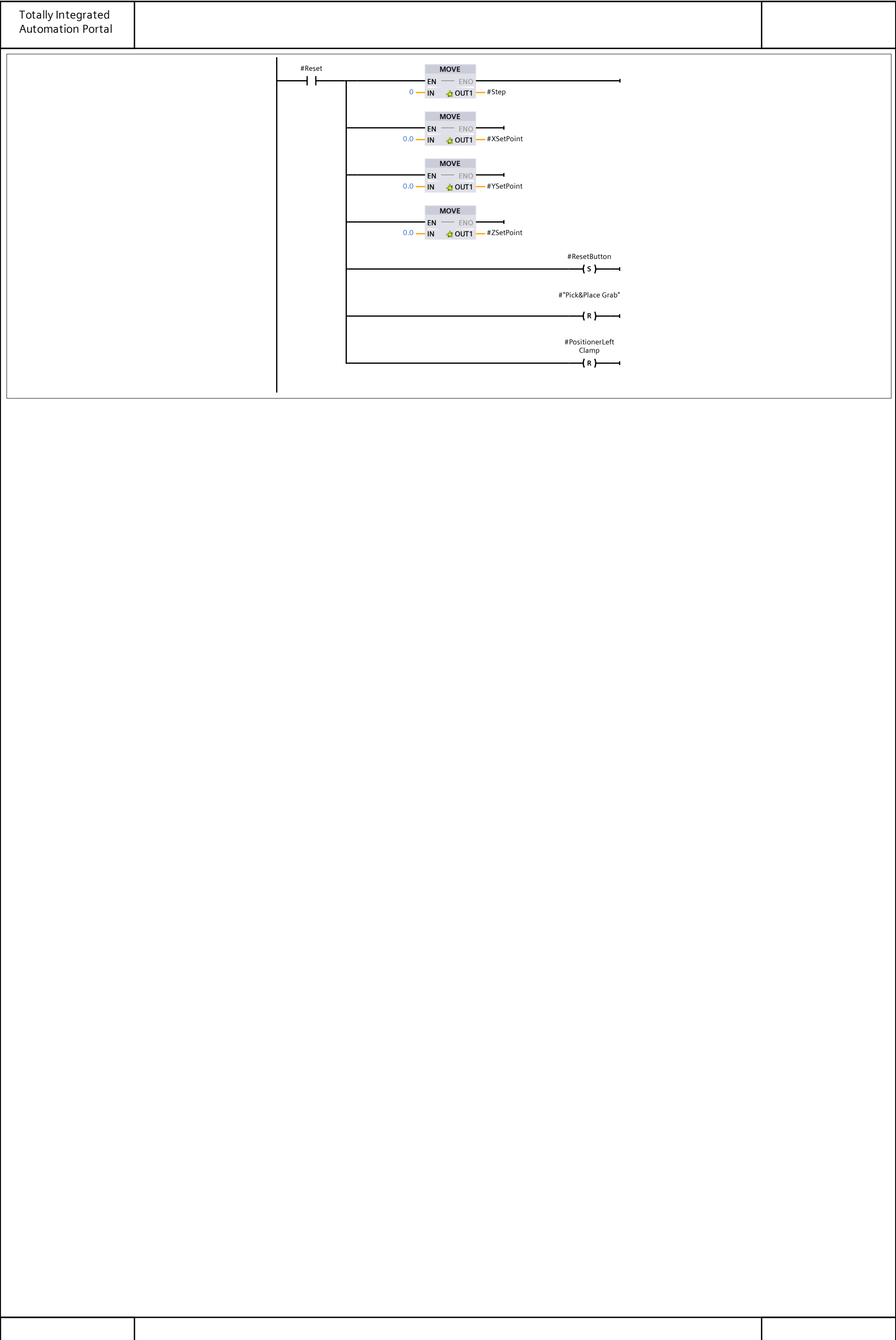
Network 22:



Network 23:



Network 24:



Totally Integrated Automation Portal

Stacker Crane [FB8]

Stacker Crane Properties

General

Name	Stacker Crane	Number	8	Type	FB	Language	LAD
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
Start_Load_Sequence	Bool	false	Non-retain
Start_UnLoad_Sequence	Bool	false	Non-retain
At_Right	Bool	false	Non-retain
At_Left	Bool	false	Non-retain
At_Middle	Bool	false	Non-retain
Moving_X	Bool	false	Non-retain
Moving_Z	Bool	false	Non-retain
Manual_Mode	Bool	false	Non-retain
▼ Output			
Load_Sequence_Done	Bool	false	Non-retain
Forks_Right	Bool	false	Non-retain
Lift	Bool	false	Non-retain
Target_Position	Int	0	Non-retain
Forks_Left	Bool	false	Non-retain
Unload_Sequence_Done	Bool	false	Non-retain
FreeCount	Int	0	Non-retain
Stored_Shelf_Percent	Int	0	Non-retain
InOut			
▼ Static			
Destination_Slot	Int	1	Non-retain
Target_Position_1	Bool	false	Non-retain
Manual_Mode_1	Bool	false	Non-retain
Stored_Shelf_Value	Int	1	Non-retain
Selected_Position	Int	1	Non-retain
Shelf_Occupied	Array[1..55] of Bool		Retain
Unload_Sequence_Active	Bool	false	Non-retain
Step	Int	0	Non-retain
Load_Sequence_Active	Bool	false	Non-retain
IEC_Counter_0_Instance	CTU_INT		Retain
orice	Bool	false	Non-retain
orice1	Bool	false	Non-retain
orice10	Bool	false	Non-retain
orice2	Bool	false	Non-retain
orice3	Bool	false	Non-retain
orice4	Bool	false	Non-retain
pulse0	Bool	false	Non-retain
IEC_Timer_0_Instance	TON_TIME		Non-retain
orice5	Bool	false	Non-retain
orice11	Bool	false	Non-retain
orice6	Bool	false	Non-retain
orice7	Bool	false	Non-retain
orice8	Bool	false	Non-retain
orice9	Bool	false	Non-retain
pulse1	Bool	false	Non-retain
trig_5	Bool	false	Non-retain
orice12	Bool	false	Non-retain
Unload_Shelf_Value	Int	1	Non-retain
▼ Temp			
Temp_Ocupate	Int		
i	Int		
Constant			

Network 1:

0001

0002 IF NOT #Manual_Mode_1 AND #Step = 0

0003 AND NOT #Load_Sequence_Active

0004 AND NOT #Unload_Sequence_Active THEN

0005

0006 #Stored_Shelf_Value := 1;

0007 WHILE (#Stored_Shelf_Value <= 55) DO

0008 IF #Shelf_Occupied[#Stored_Shelf_Value] = FALSE THEN

0009 EXIT;

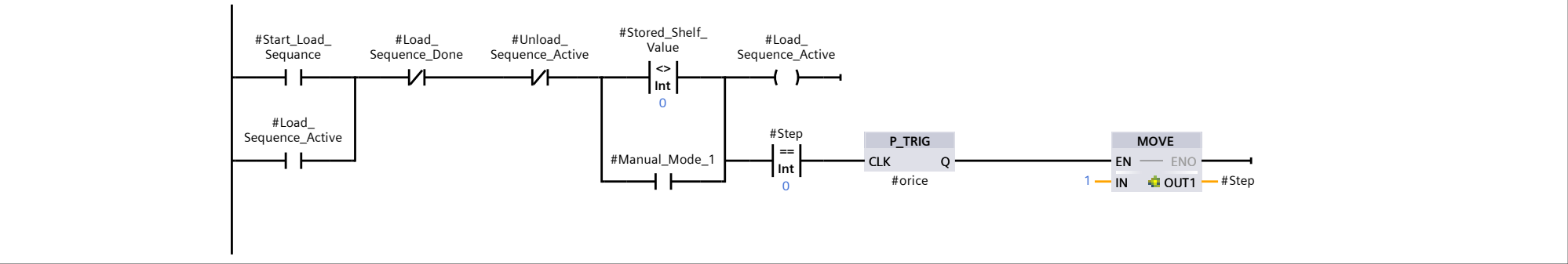
0010 END_IF;

0011 #Stored_Shelf_Value := #Stored_Shelf_Value + 1;

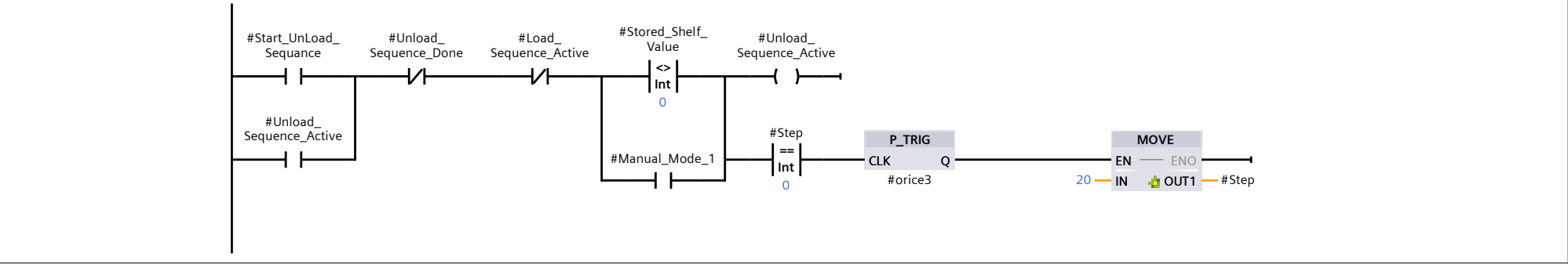
0012 END_WHILE;

```
0013
0014     IF #Stored_Shelf_Value > 55 THEN
0015         #Stored_Shelf_Value := 0;
0016     ELSE
0017         #Destination_Slot := #Stored_Shelf_Value;
0018     END_IF;
0019
0020 END_IF;
```

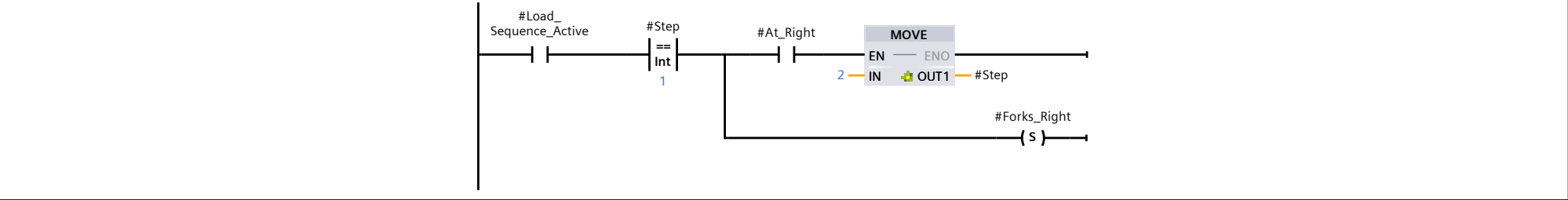
Network 2: Step 0 - Load Sequence Active



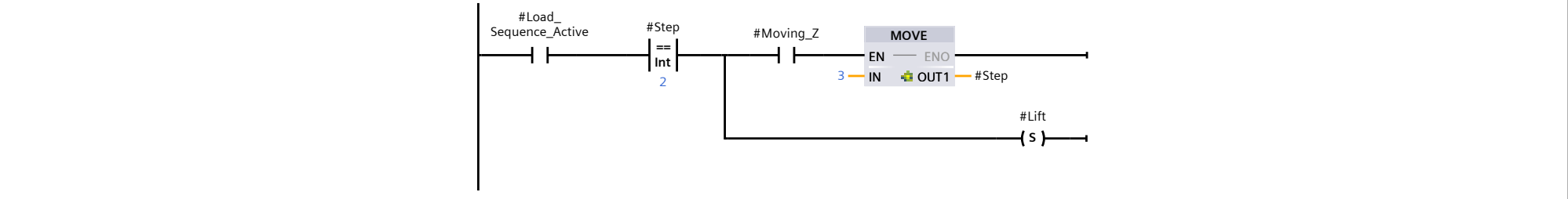
Network 3: Step 0 - UnLoad Sequence Active



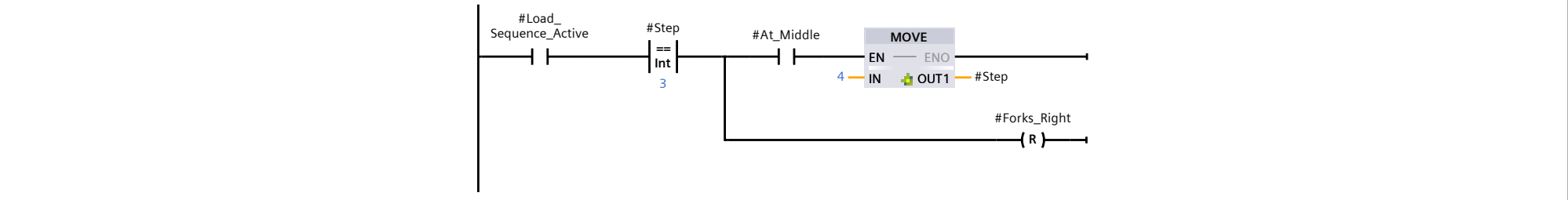
Network 4: Step 1 Forks Right



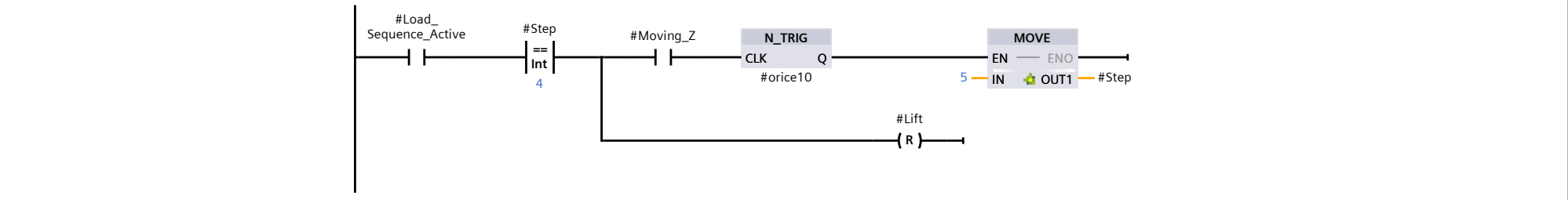
Network 5: Step 2 Lift up



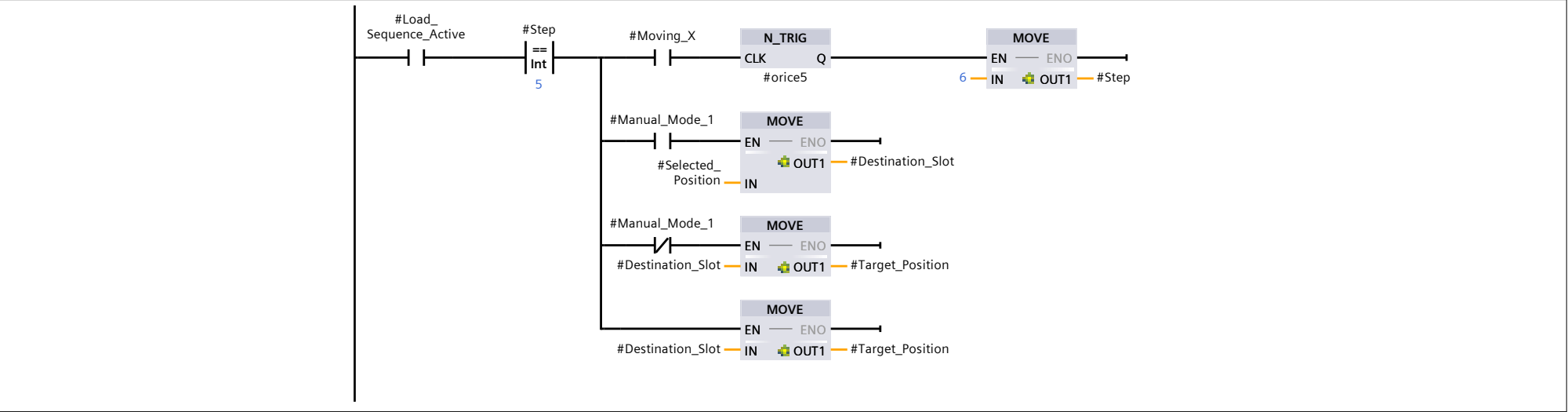
Network 6:



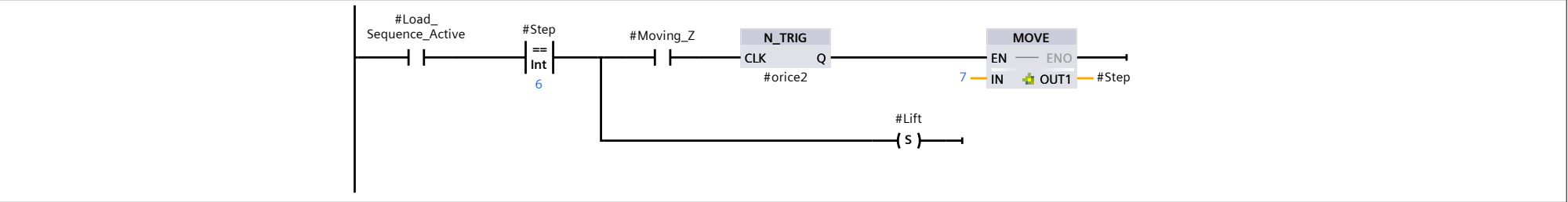
Network 7:



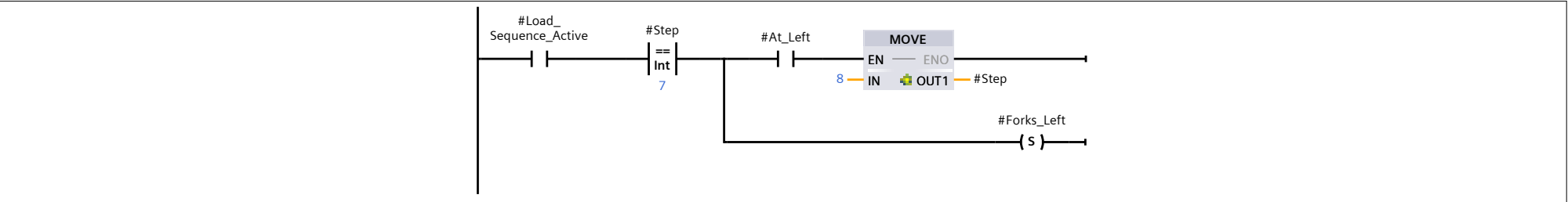
Network 8:



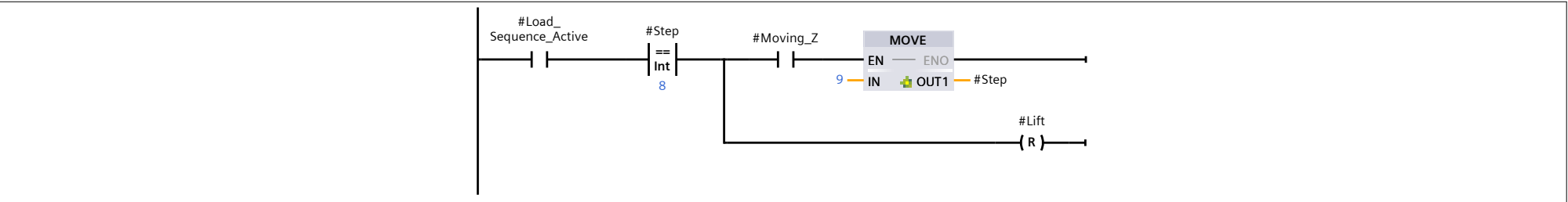
Network 9:



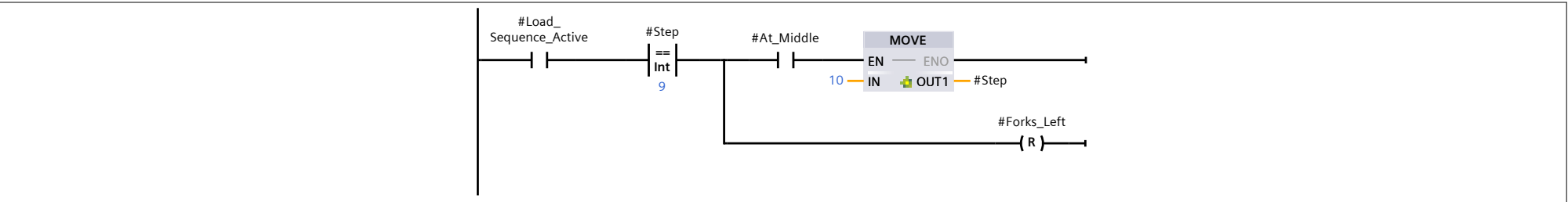
Network 10:



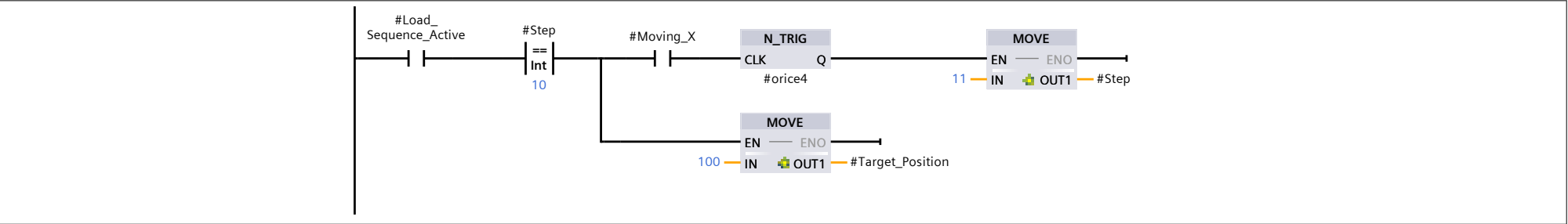
Network 11:



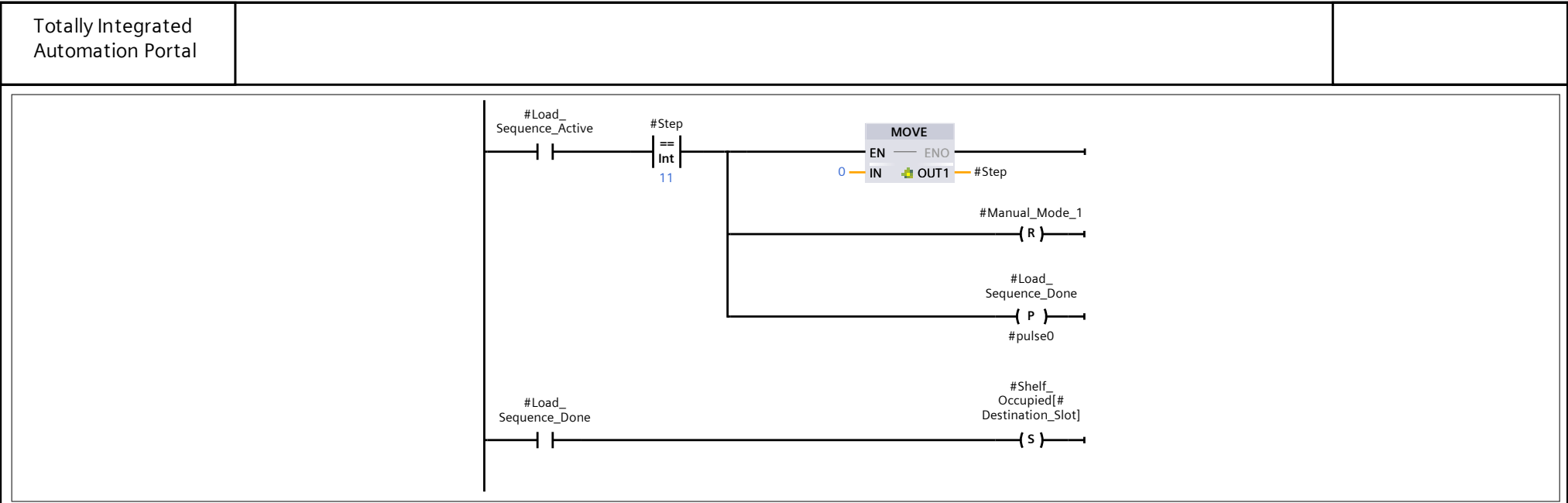
Network 12:



Network 13:



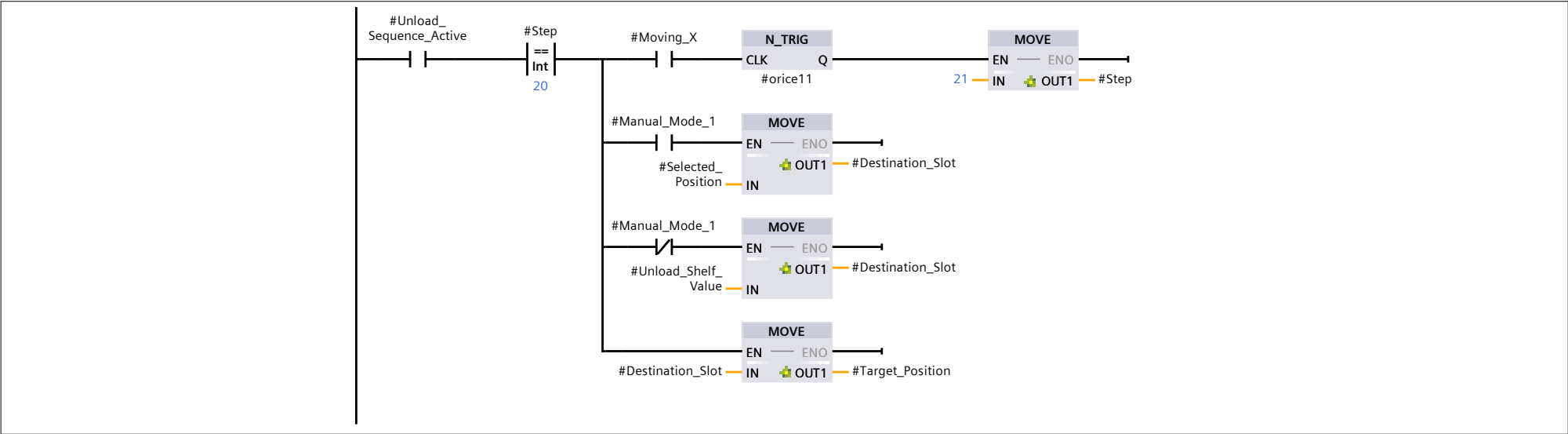
Network 14:



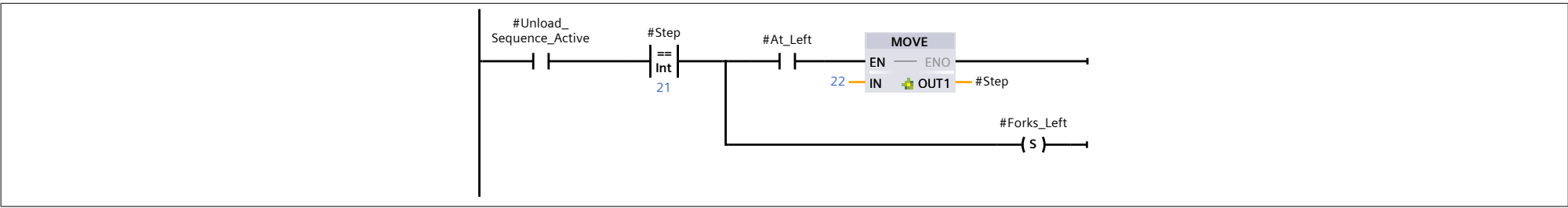
Network 15:

```
0001 // Adaugă aceeași protecție ca la Network 1
0002 IF NOT #Manual_Mode_1 AND #Step = 0
0003     AND NOT #Load_Sequence_Active
0004     AND NOT #Unload_Sequence_Active THEN
0005
0006     #Unload_Shelf_Value := 55;
0007     WHILE (#Unload_Shelf_Value >= 1) DO
0008         IF #Shelf_Occupied[#Unload_Shelf_Value] = TRUE THEN
0009             EXIT;
0010         END_IF;
0011         #Unload_Shelf_Value := #Unload_Shelf_Value - 1;
0012     END_WHILE;
0013
0014     IF #Unload_Shelf_Value < 1 THEN
0015         #Unload_Shelf_Value := 0;
0016     END_IF;
0017
0018 END_IF;
```

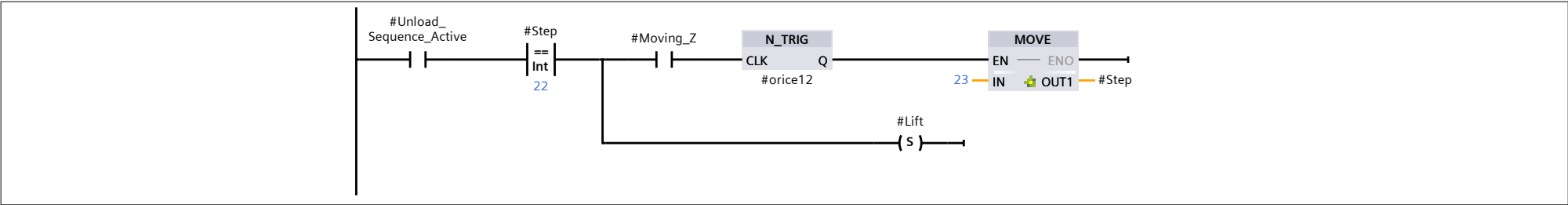
Network 16: Step 20 - Move to target position / Move X position



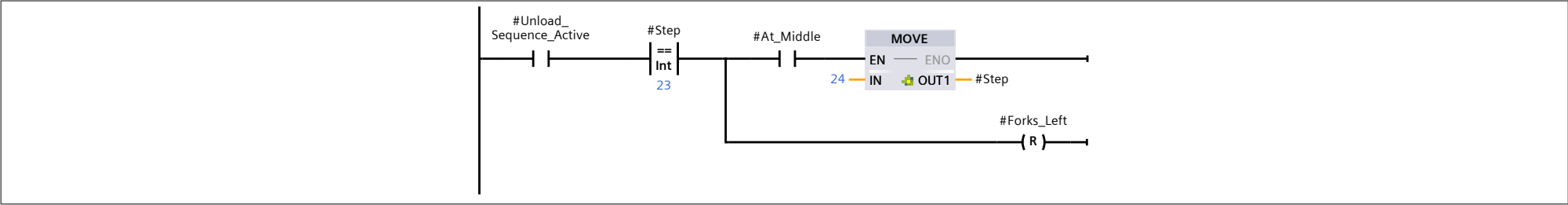
Network 17:



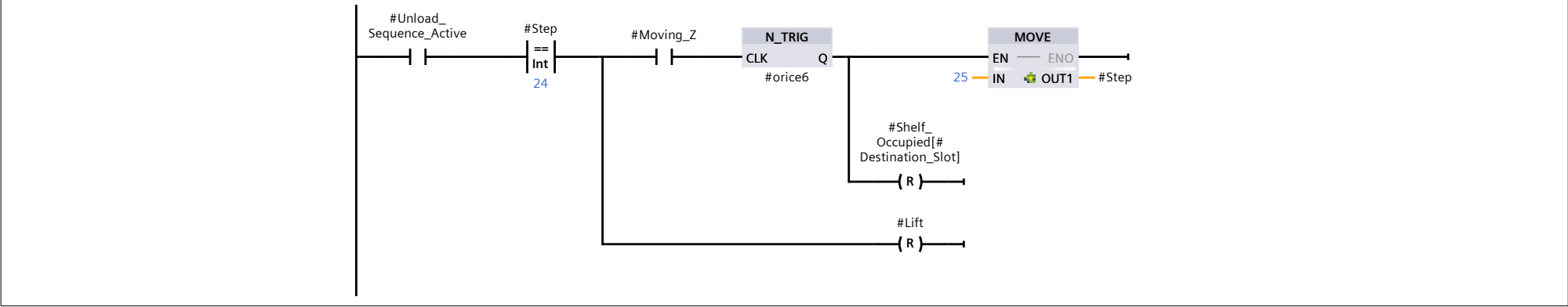
Network 18:



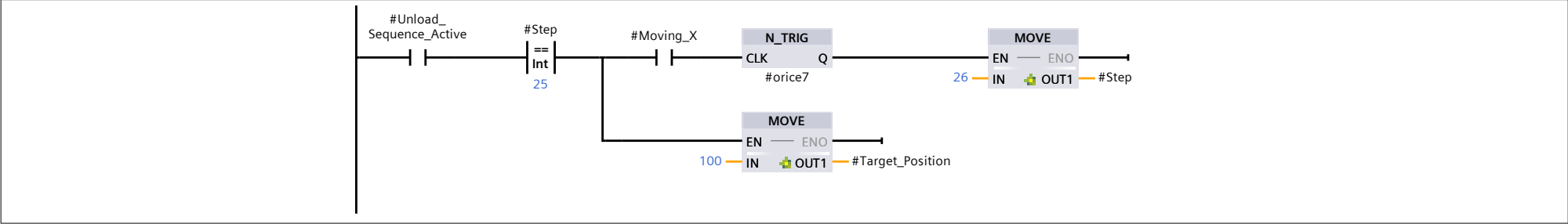
Network 19:



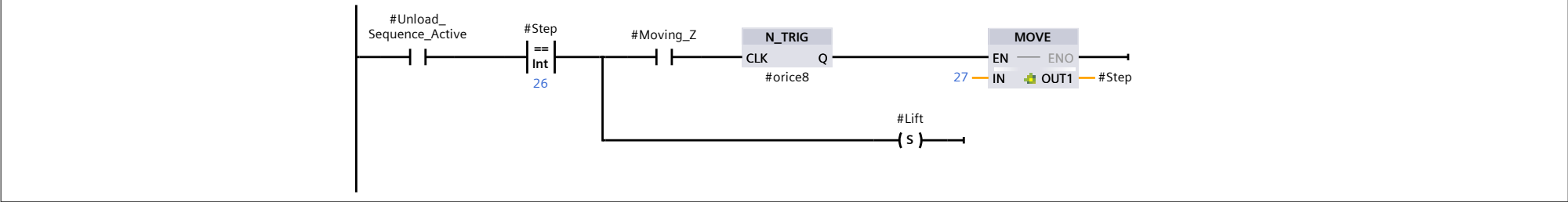
Network 20:



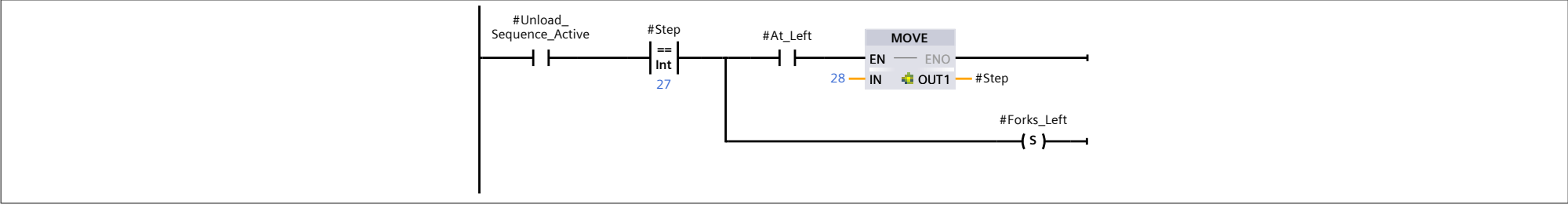
Network 21:



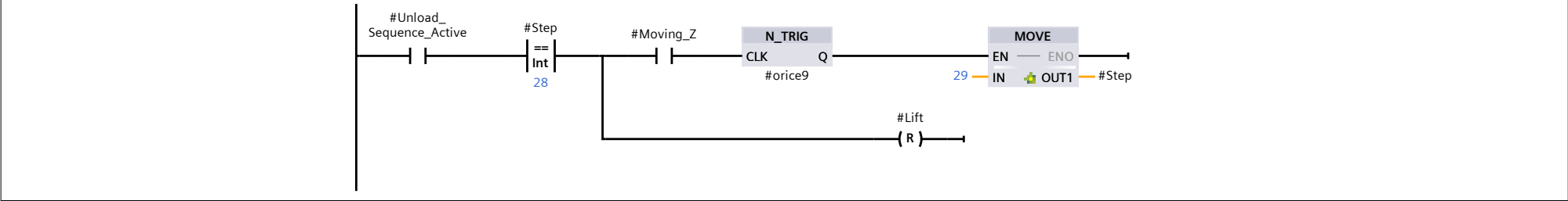
Network 22:



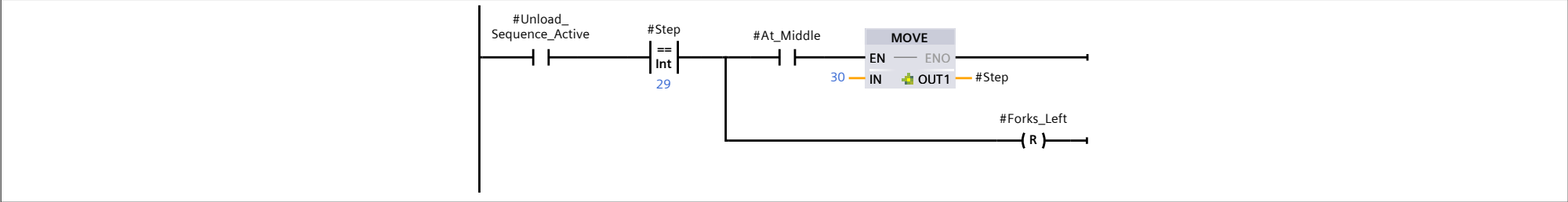
Network 23:



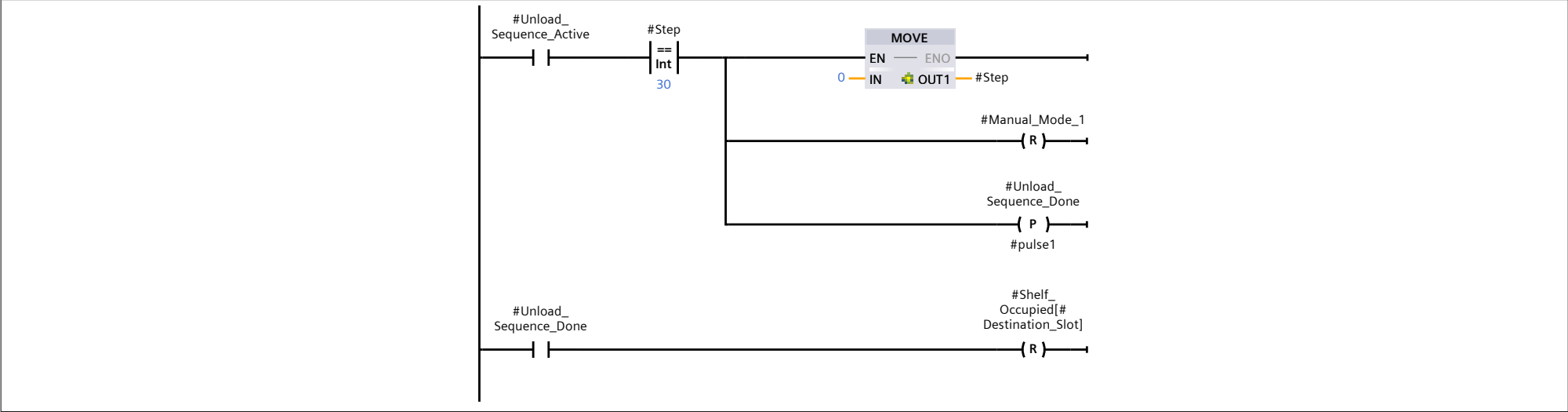
Network 24:



Network 25:



Network 26:



Network 28:

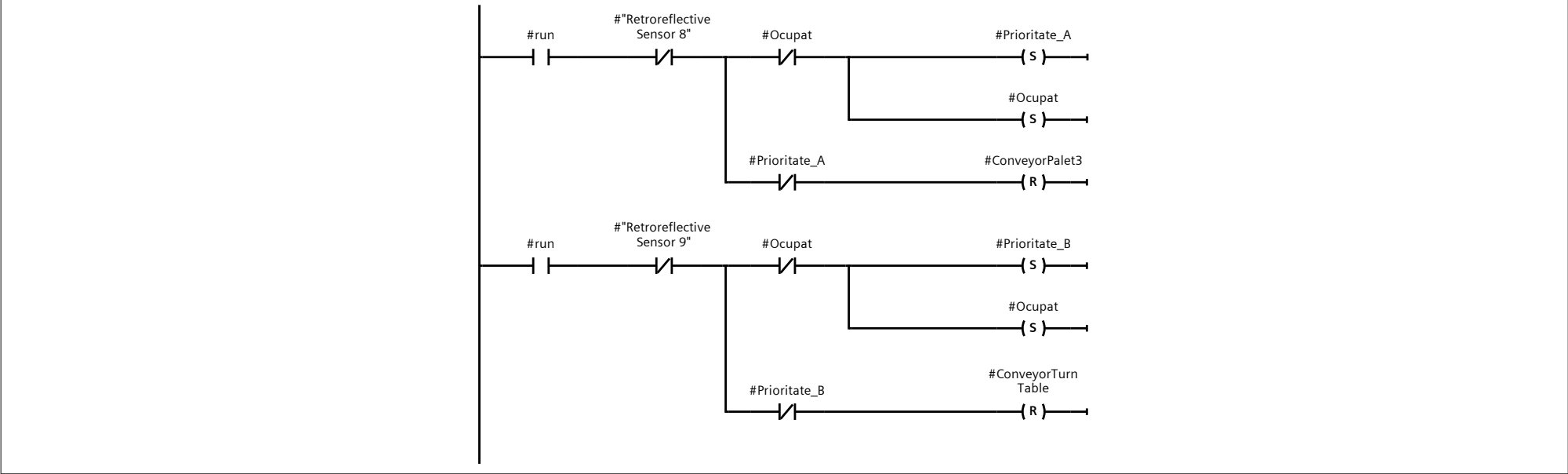
```
0001 // 1. Resetam contorul temporar la fiecare ciclu de scanare
0002 #Temp_Ocupate := 0;
0003
0004 // 2. Numaram cate rafturi sunt ocupate fizic (verificam array-ul tau)
0005 FOR #i := 1 TO 54 DO
0006     IF #Shelf_Occupied[#i] = TRUE THEN
0007         #Temp_Ocupate := #Temp_Ocupate + 1;
0008     END_IF;
0009 END_FOR;
0010
0011 // 3. Calculam locurile neocupate (scriem direct in Output-ul tau)
0012 #FreeCount := 54 - #Temp_Ocupate;
0013
0014 // Protectie limite (sa nu scada sub 0)
0015 IF #FreeCount < 0 THEN
0016     #FreeCount := 0;
0017 END_IF;
0018
0019 // 4. Calculam procentul de ocupare (scriem direct in Output-ul tau)
0020 #Stored_Shelf_Percent := (#Temp_Ocupate * 100) / 54;
0021
0022 // Protectie limite procentaj
0023 IF #Stored_Shelf_Percent > 100 THEN
0024     #Stored_Shelf_Percent := 100;
0025 ELSIF #Stored_Shelf_Percent < 0 THEN
0026     #Stored_Shelf_Percent := 0;
0027 END_IF;
```

TurnTable [FB7]

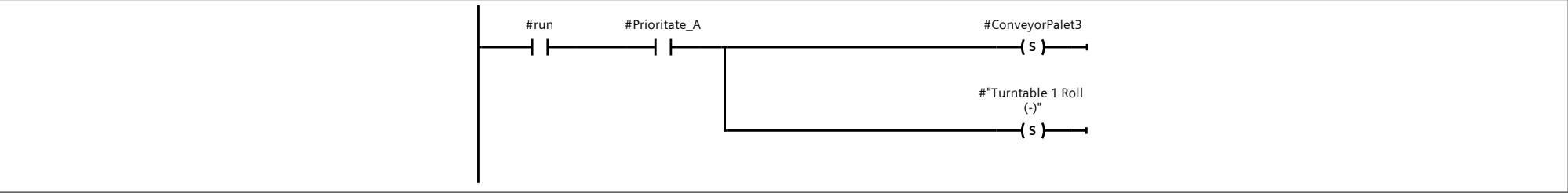
TurnTable Properties							
General							
Name	TurnTable	Number	7	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
SensorTurnTable1	Bool	false	Non-retain
Turntable 0 (Back Limit)_1	Bool	false	Non-retain
Retroreflective Sensor 9	Bool	false	Non-retain
Retroreflective Sensor 8	Bool	false	Non-retain
Turntable 1 (Back Limit)	Bool	false	Non-retain
Retroreflective Sensor 11	Bool	false	Non-retain
Turntable 0 (Limit 90)	Bool	false	Non-retain
Turntable 1 (Front Limit)	Bool	false	Non-retain
Turntable 1 (Limit 90)	Bool	false	Non-retain
Turntable 1 (Limit 0)	Bool	false	Non-retain
Retroreflective Sensor 16	Bool	false	Non-retain
Start	Bool	false	Non-retain
run	Bool	false	Non-retain
▼ Output			
Turntable 0 Roll (-)	Bool	false	Non-retain
Turntable 0 Turn	Bool	false	Non-retain
ConveyorTurnTable	Bool	false	Non-retain
Turntable 0 Roll (+)	Bool	false	Non-retain
Turntable 1 Roll (-)	Bool	false	Non-retain
Turntable 1 Turn	Bool	false	Non-retain
ConveyorPalet3	Bool	false	Non-retain
Turntable 1 Roll (+)	Bool	false	Non-retain
InOut			
▼ Static			
Ocupat	Bool	false	Non-retain
Prioritate_B	Bool	false	Non-retain
Prioritate_A	Bool	false	Non-retain
IEC_Timer_0_Instance	TP_TIME		Non-retain
Orice	Bool	false	Non-retain
IEC_Timer_0_Instance_1	TON_TIME		Non-retain
orice1	Bool	false	Non-retain
IEC_Timer_0_Instance_2	TON_TIME		Non-retain
IEC_Timer_0_Instance_3	TON_TIME		Non-retain
IEC_Timer_0_Instance_4	TP_TIME		Non-retain
IEC_Timer_0_Instance_5	TON_TIME		Non-retain
Orice2	Bool	false	Non-retain
IEC_Timer_0_Instance_6	TP_TIME		Non-retain
Step	Int	0	Non-retain
IEC_Timer_0_Instance_7	TON_TIME		Non-retain
IEC_Timer_0_Instance_8	TON_TIME		Non-retain
▼ Temp			
mem1	Bool		
Constant			

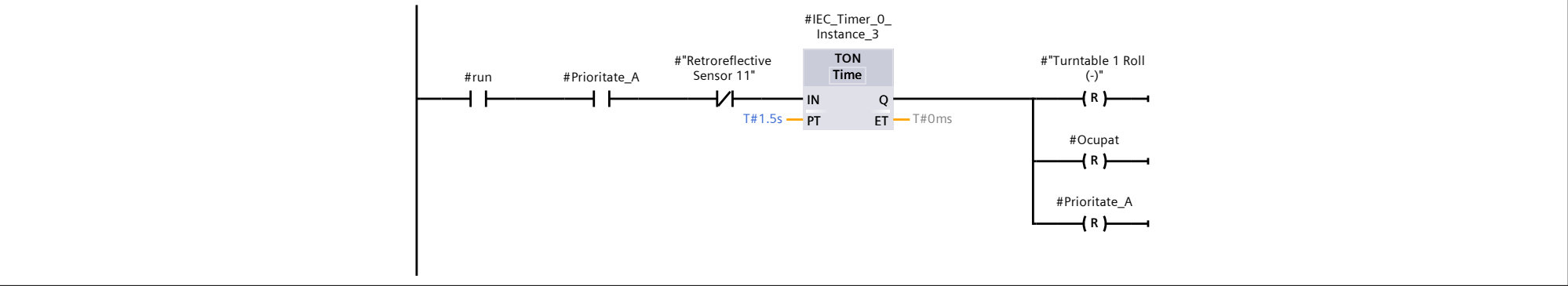
Network 1:



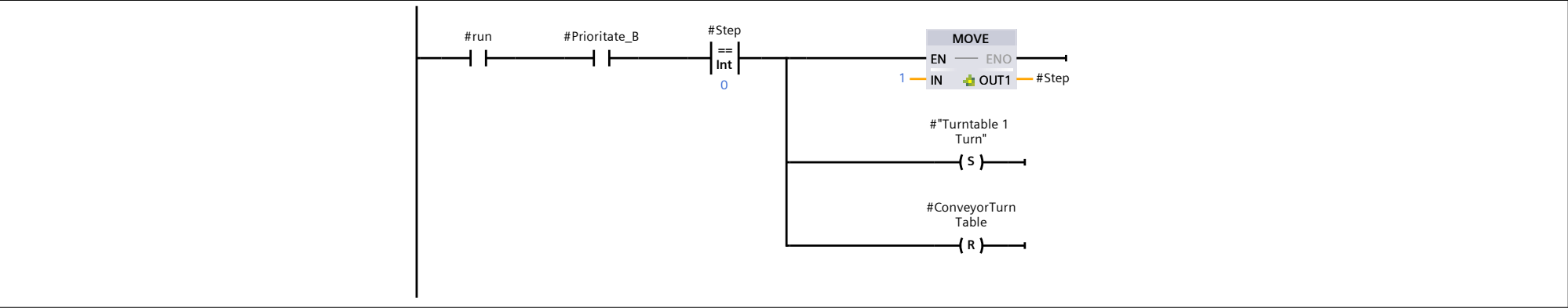
Network 2:



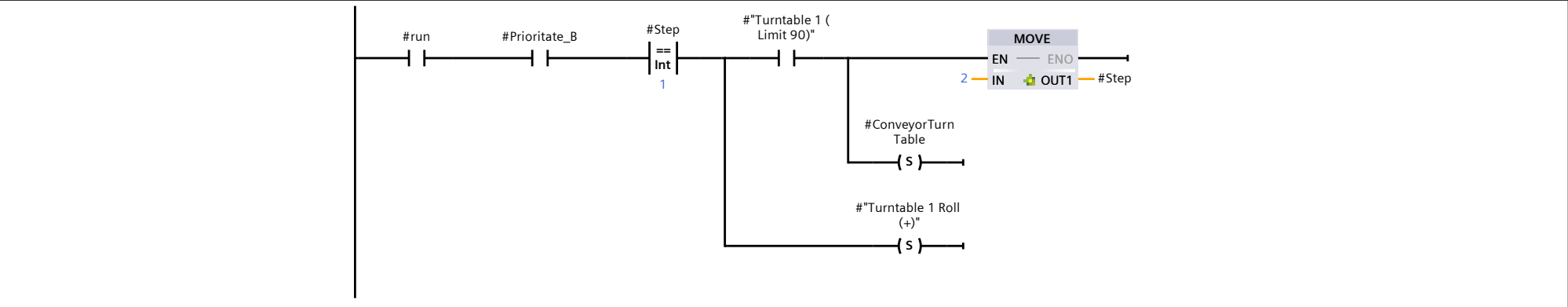
Network 3:



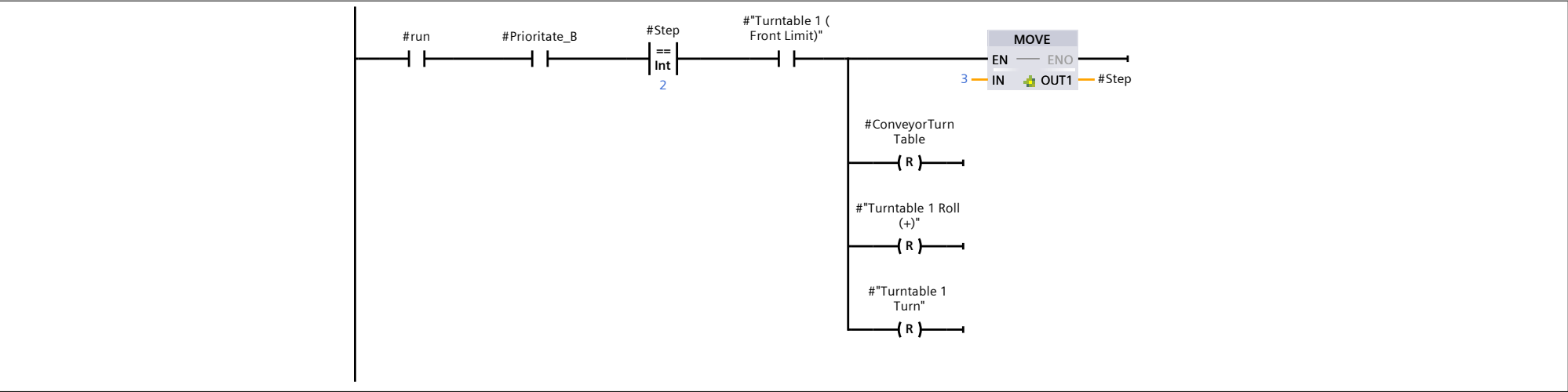
Network 4:



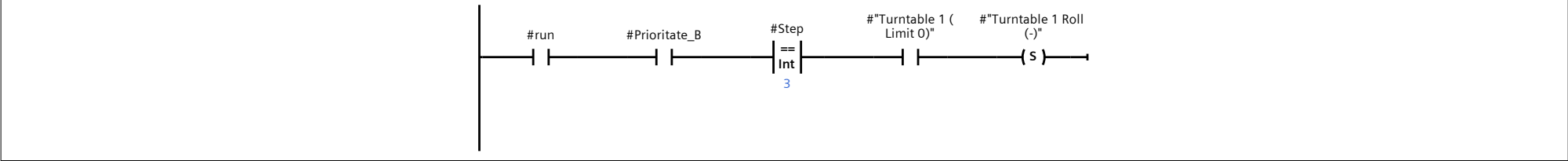
Network 5:



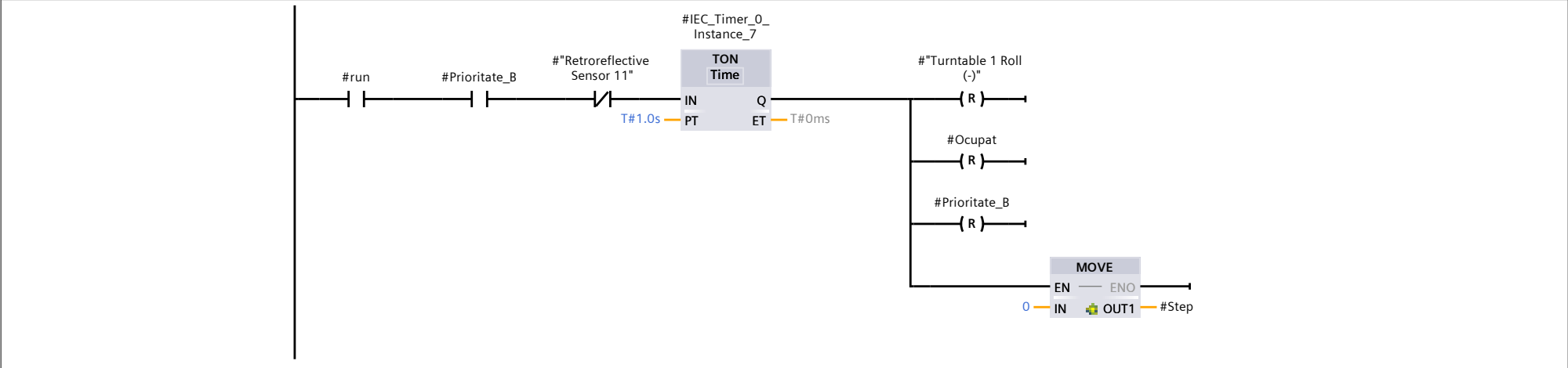
Network 6:



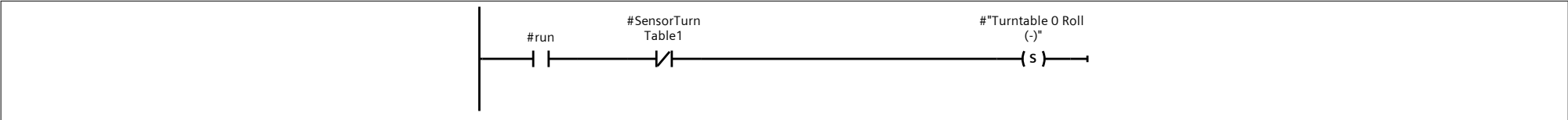
Network 7:



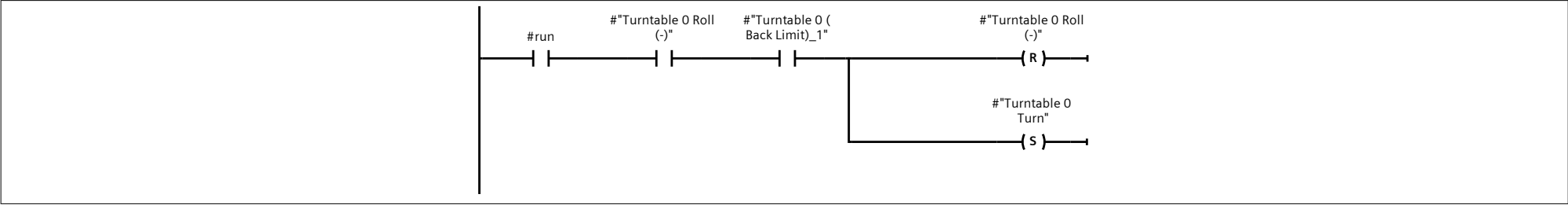
Network 8:



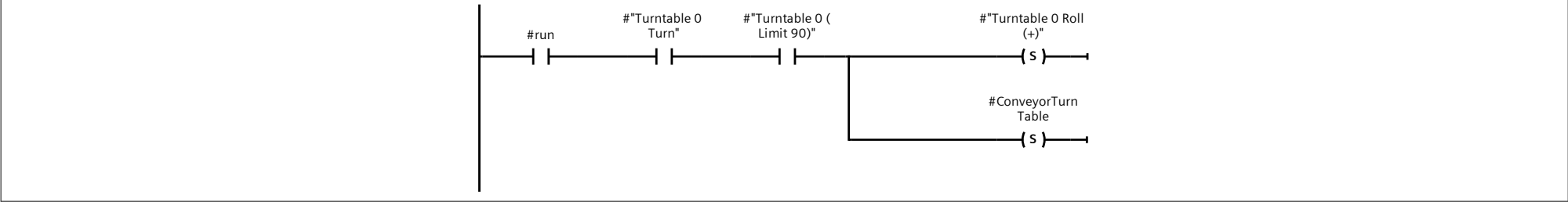
Network 9:



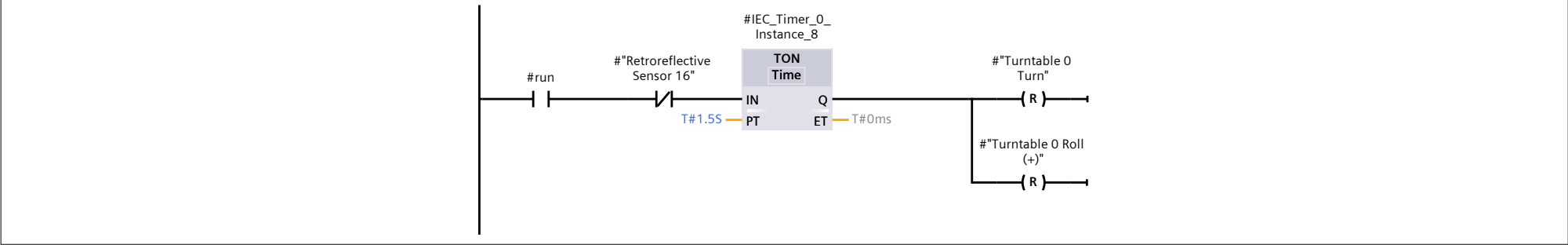
Network 10:



Network 11:



Network 12:

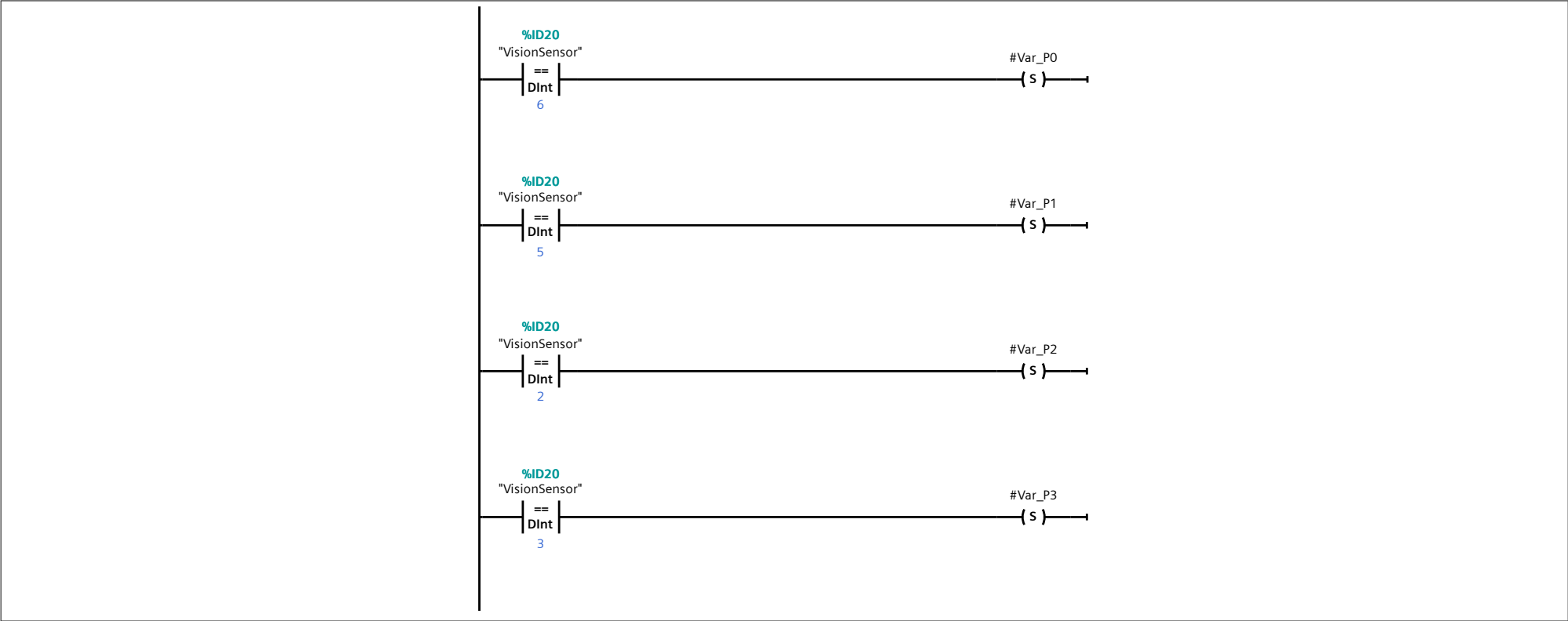


VisionSorter [FB3]

VisionSorter Properties							
General							
Name	VisionSorter	Number	3	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

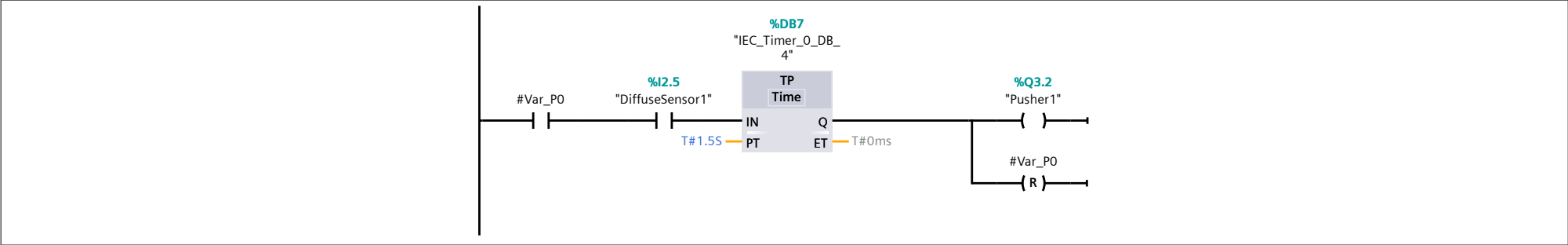
Name	Data type	Default value	Retain
Input			
Output			
InOut			
▼ Static			
Var_P0	Bool	false	Non-retain
Var_P1	Bool	false	Non-retain
Var_P2	Bool	false	Non-retain
Var_P3	Bool	false	Non-retain
Temp			
Constant			

Network 1:



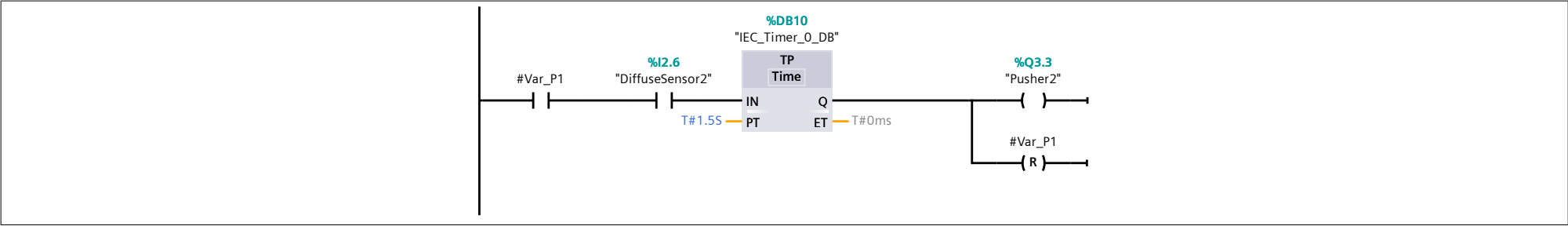
Network 2:

Pusher1

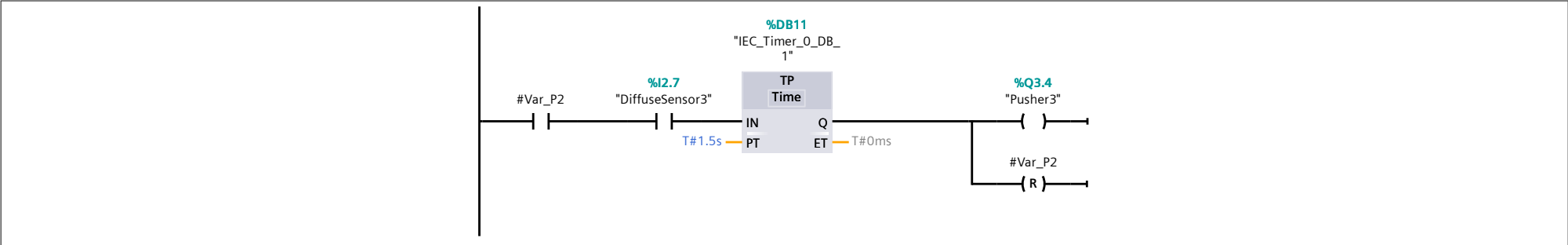


Network 3:

Pusher2



Network 4:



Network 5:

